Distributed Policy Optimization under Partial Observability: Tractability, Linear Speedup, and Communication Efficiency

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Abstract

We consider the problem of policy optimization within the context of Partially Observable Markov Decision Processes (POMDPs) in a distributed setting where M clients collaborate under the coordination of a central server. We develop a new theoretical framework to characterize the performance of distributed policy optimization for POMDPs. Specifically, we propose a novel actor-critic framework, where the agents collaboratively perform policy improvement and evaluation to maximize the benefit of parallelization. Under linear function approximation, we rigorously establish global convergence rates for a wide range of softmax policies, achieving linear speedup in sample complexity along with sublinear communication complexity. Our analysis builds on a novel, sharp bound for local drift terms and a parameter stacking technique to analyze temporal correlations that are of independent interest in studying POMDPs.

1 Introduction

Reinforcement learning (RL) is a sequential decision-making process where an agent aims to maximize her cumulative reward through repeated interactions with an unknown environment (Sutton and Barto, 2018). RL is described by the Markov Decision Process (MDP) (Puterman, 1994), in which the agent chooses an action at each moment based on the current state and receives a reward. The environment then transitions to a new state in response to this action. An essential assumption in MDPs is that the agent has complete knowledge of the environment's state at every moment. However, in many practical applications, such as predicting stock prices (Kabbani and Duman, 2022) and training humanoid robots to walk (Lauri et al., 2022), the agent has access only to unreliable or incomplete information about the environment's state. We model these sequential decision-making problems as Partially Observable Markov Decision Processes (POMDPs) (Monahan, 1982). In a POMDP, unlike an MDP, the agent takes action based solely on noisy *observations* of the states.

While POMDPs provide significant modeling flexibility, theoretical investigations into their tractability present a discouraging outlook. (Papadimitriou and Tsitsiklis, 1987) and (Jin et al., 2020a) have established that obtaining the optimal policy for POMDPs using exact methods, such as dynamic programming, is computationally and statistically intractable. Furthermore, existing RL theories for POMDPs often emphasize learning the transition dynamics while assuming access to a perfect planning algorithm, whose computational complexity scales exponentially with the observation space (Guo et al., 2023a; Jin et al., 2020a; Liu et al., 2022a). Despite these discouraging theoretical results, POMDP algorithms based on deep reinforcement learning have achieved tremendous empirical success (Ma et al., 2024; Zhuang et al., 2024). The adoption of neural networks enables the extraction of structured representations from historical observations, which is advantageous for policy optimization (albeit at the cost of a small approximation error) (Esslinger et al., 2022). More importantly, empirical methods have also demonstrated the benefits of parallelization in the policy optimization of POMDPs (Cui et al., 2024; Rudin et al., 2022; Thibault et al., 2024) (see Fig 1 for an illustration).

This work takes the first step to bridge this gap between theory and practice by developing new theoretical results that aim to explain the empirical success of POMDPs, particularly in the parallel setting.

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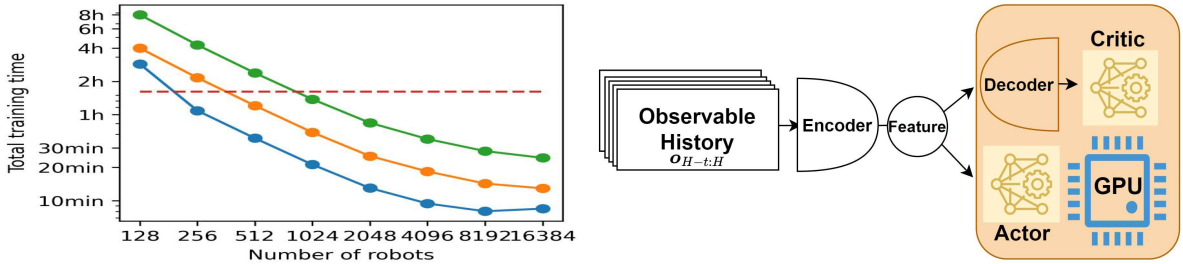


Figure 1: The left figure, adapted from Figure 4(b) in (Rudin et al., 2022), demonstrates that partially observable RL is consistently accelerated when more agents are trained in parallel, even though it involves greater complexity than MDPs. This phenomenon is what our theory seeks to explain. The right figure illustrates practical solutions for POMDPs, where actor and critic networks carry out massively parallel policy optimization after processing feature representations extracted from historical inputs. The colored region highlights the procedure we aim to analyze.

1.1 Contributions

We study the problem of distributed reinforcement learning in partially observable environments, where multiple agents collaboratively learn the optimal policy of a shared, underlying POMDP. We show why policy optimization for POMDP benefits from data parallelism, similar to MDP, and where the fundamental differences lie in their convergence rates. Our contributions include:

- *Algorithm:* We propose algorithms to model practical policy evaluation and improvement for POMDPs, preserving key ingredients of distributed optimization, such as client-server communication, multiple local steps, stochastic gradients, and weighted averaging.
- *Complexity:* We establish global convergence rates for a wide range of softmax policies. Under linear function approximation, we show that our algorithms enjoy linear speedup in sample complexity and a communication complexity that is sublinear in the computation cost per agent. These terms, as well as the computation complexity, are independent of the sizes of the state and observation space.
- *Technique:* We offer new methods to analyze distributed POMDP training from a theoretical standpoint and a technique to bound the local drift effect with Riemannian integration.

1.2 Related Work

Partially-observable RL Existing POMDP theories primarily focus on learning transition dynamics for subclasses of POMDPs with polynomial sample complexity (Cai et al., 2022; Liu et al., 2022b). Such studies usually assume access to ideal planning algorithms (Jin et al., 2020a; Lee et al., 2023), which outputs the value function and the corresponding optimal policy, given estimated transition kernels. While few works implement the planning oracles in detail (Golowich et al., 2022a; Guo et al., 2023b), their methods involve an exhaustive search over all the possible observation histories, demanding computation and memory resources that scale with the observation space size raised to the power of the horizon length. Recent studies (Cayci and Eryilmaz, 2024; Zhang et al., 2024) improve the tractability of previous work with policy optimization method but failed to explain how it can be accelerated by parallel training. We provide a finite-time sample complexity analysis on the error in the planning procedure, whose complexity is irrelevant to the size of the observation space. We also quantitatively characterize how our algorithm benefits from massive parallel training, aligning with the empirical phenomenon observed in (Rudin et al., 2022).

Distributed Policy Optimization. Recent distributed policy gradient methods mainly focus on Markov policies (Wang et al., 2024) with tabular softmax parameterization (Yang et al., 2024). Their data parallel routine synchronizes every iteration (Lan et al., 2023), resulting in high communication costs. We extend previous results to history-dependent policies with general softmax parameterization and allow multiple local steps between updates. Many existing works on distributed TD learning did not contain a linear speedup analysis in their sample complexity (Liu and Olshevsky, 2021). We extend recent TD learning studies (Tian et al., 2024; Wang et al., 2023) from MDPs to POMDPs, ensure sublinear communication complexity, and relax the condition for achieving linear speedup in sample complexities, which is crucial to massively parallel training.

2 Problem Formulation

Notations. We use $\lesssim$ or $\widetilde{O}$ to omit constants or logarithmic factors in the expressions, respectively.

For $p \in [1, \infty]$ and $v, w \in \mathbb{R}^d$, $\|v\|_p$ denotes the p-norm of the vector v and $\langle v, w \rangle_{\mathbb{R}^d}$ denotes the inner product between the vectors v and w. We use $\Delta(\mathcal{X})$ to denote the probability simplex over a finite set $\mathcal{X}$.For matrices A and B, the notation $A \succeq B$ implies $A - B$ is positive semi-definite. We denote the Moore-Penrose pseudo-inverse of a matrix A using $A^\dagger$. Lastly, for probability distributions P and Q, $D_{KL}(P||Q) = \mathbb{E}_{X \sim P}\left[\log \frac{p(X)}{q(X)}\right]$ denotes the Kullback-Leibler (KL) divergence between P and Q.

POMDP. We consider an episodic Partially Observable Markov Decision Process (POMDP) with state space $\mathcal{S}$, action space $\mathcal{A}$, observation space $\mathcal{O}$, and discount factor $\gamma \in (0, 1)$ over a finite horizon H, the action space is finite but state and observation spaces can be continuous. A POMDP has a reward function $r_h : \mathcal{S} \times \mathcal{A} \to [0, 1]$, transition kernel $\mathbb{T}_{h,a}(\cdot|s)$, and emission matrix $\mathbb{O}_h(\cdot|s)$. At time h, state s_h transitions to $s_{h+1} \sim \mathbb{T}_{h,a_h}(\cdot|s_h)$ with reward $r_h(s_h, a_h)$. The agent observes $o_{h+1} \sim \mathbb{O}_{h+1}(\cdot|s_{h+1})$ to take action a_{h+1}. We draw the initial state S_0 from a distribution $\rho \in \Delta(\mathcal{S})$. We denote $Z_h := (O_0, A_0, \ldots, O_{h-1}, A_{h-1}, O_h)$ as the observable history in a history space $\mathcal{Z}_h$. A policy π consists of mappings $\pi_h : \mathcal{Z}_h \to \Delta(\mathcal{A})$, where $\pi_h(a_h|Z_h)$ is the probability of action a_h given history Z_h. For notation brevity, we denote $\bar{z}_h$ as (z_h, a_h). Like MDPs, we associate any policy π with a value function and a Q-function:

$$\mathsf{V}_t^\pi(z_t) := \mathbb{E}^\pi\left[\sum_{h=t}^{H} \gamma^{h-t} r_h(S_h, A_h) \big| Z_t = z_t\right]$$

$$\mathsf{Q}_t^\pi(z_t, a_t) := \mathbb{E}^\pi\left[\sum_{h=t}^{H} \gamma^{h-t} r_h(S_h, A_h) \big| \bar{Z}_t = (z_t, a_t)\right]$$

We also define an advantage function associated with a policy π, which is given by $\mathsf{A}_t^\pi(z_t, a_t) := \mathsf{Q}_t^\pi(z_t, a_t) - \mathsf{V}_t^\pi(z_t)$. The learning objective is to find the policy π from a given class of policies Π that maximizes the expected cumulative reward, i.e.,

$$\sup_{\pi \in \Pi} \mathbb{E}^\pi\left[\sum_{h=0}^{H-1} \gamma^h r_h(S_h, A_h)\right]. \tag{1}$$

Policy Parametrization. For the convenience of optimization over a large observation space, we associate the policies with a parameter $\theta \in \mathbb{R}^{d_\theta}$. When the action space is finite, we optimize the policies within the class of history-dependent softmax policies (Agarwal et al., 2019), which take the form of

$$\pi_h^\theta(a_h|z_h) := \frac{\exp f_\theta(z_h, a_h; h)}{\sum_{a_h \in \mathcal{A}} \exp f_\theta(z_h, a_h; h)} \tag{2}$$

Here, $f_\theta(\cdot, \cdot; h) : \mathcal{Z}_h \times \mathcal{A} \to \mathbb{R}$ is a real-valued function for any time index h. Eq. (2) encompasses a wide range of policies, including "tabular softmax policies", where $f_\theta(z_h, a_h; h) = \theta(z_h, a_h, h)$; "log-linear policies," where $f_\theta(z_h, a_h; h) = \theta^\top \xi_h(z_h, a_h)$ and $\xi_h(z_h, a_h)$ is a known feature; and "neural softmax policies," where $f_\theta(\cdot, \cdot; \cdot)$ is a non-linear sequential neural network.

Linear Function Approximation. Following the convention of theoretical RL (Bhandari et al., 2018; Tsitsiklis and Van Roy, 1996), we adopt linear function approximation for the Q-functions in the face of a continuous observation space. We assume that the Q-function of any policy π can be represented by a set of vectors $\{\psi_h^\pi\}_{h=0}^{H-1}$ as follows: $\mathsf{Q}_h^\pi(z_h, a_h) = \langle \phi_h(z_h, a_h), \psi_h^\pi \rangle$ for all $(z_h, a_h) \in \mathcal{Z}_h \times \mathcal{A}$ and $h \in \{0, 1, \ldots, H-1\}$. Here, vectors $\{\psi_h^\pi\}_{h=1}^{H}$ encode the information of the policy, while $\{\phi_h\}_{h=0}^{H-1}$ denotes a collection of known functions that form a compact representation of the POMDP dynamics at time h. Rich subsets of POMDPs satisfy this assumption, including tabular POMDPs, where value functions are the inner product between the α-vectors and belief states (Pineau et al., 2006) with $d = |\mathcal{S}|$; L-decodable POMDPs with rank d_{rank} transitions (Guo et al., 2023b), where $Q_h(z_h, a_h) = p_h^\top \omega_h^\pi$ and $d = d_{\text{rank}}$[1]

Remark 1. We assume the features are known and primarily focus on policy optimization. The problem of learning good feature representation has been studied for various subclasses of POMDPs (Guo et al., 2023b; Uehara et al., 2022; Zhang et al., 2024).

Distributed Optimization We consider reinforcement learning in a distributed regime, where M agents collaboratively optimize a shared POMDP. A central server updates a shared policy for R rounds, and client agents take K local steps in between. $T = RK$ denotes the computation budget per agent.

3 Algorithm Design

In this section, we outline our proposed algorithm. Our proposed approach, called Distributed Partially Observable Natural Actor-Critic (DPAC) (Alg. 1), relies on the iterative optimization of a shared policy. In each iteration r, DPAC collaboratively evaluates the current policy using a sub-routine known as DPTD (Alg. 2). Based on this evaluation, it refines the current estimate collaboratively through another sub-routine, DPPG (Alg. 3). After R iterations, DPAC returns a uniform mixture of the

[1] see Eq. (11) in (Zhang et al., 2024), Prop 2.3 in (Jin et al., 2020b) for details.

policies parameterized by $\{\theta_0, \theta_1, \ldots, \theta_{R-1}\}$, with θ_r representing the policy parameter at the start of the rth iteration. The following sections will provide a detailed description of our policy evaluation and improvement routines.

Algorithm 1: Distributed **P**artially-observable Natural **A**ctor-**C**ritic (DPAC)

1: **Initialize** actor param.θ_0, critic param. ψ_0.
2: **for** round r in $0, 1, \ldots, R-1$ **do**
3: Send θ_r, ψ_r to client agents
4: Evaluate Q-function $\widehat{\mathsf{Q}}^{\pi^{\theta_r}}$ by running Alg 2 for T_e steps.
5: Send $\widehat{\mathsf{Q}}^{\pi^{\theta_r}}$ to client agents
6: Approximate policy gradient ω_r by running line 6 to 17 in Alg 3 with K local steps.
7: Update policy in central server $\theta_{r+1} \leftarrow \theta_r + \eta_r \cdot \omega_r$
8: **end for**
9: **return** Uniform mixture of $\theta_0, \theta_1, \ldots, \theta_{R-1}$

3.1 Policy Evaluation

Designing a distributed policy evaluation pipeline for POMDPs presents three additional challenges. First, exact evaluation of the value functions involves an exhaustive search through the history space $\mathcal{Z}_H$ to calculate a marginalization constant, bringing computational and spatial costs that scale with $(\mathcal{OA})^H$ (Zhang et al., 2024). Second, the Bellman operator for finite-horizon POMDPs is naturally time-varying, potentially resulting in H consecutive regression problems, which could result in compounding of errors (Lambert et al., 2022). Optimizing non-Markovian policies in finite horizon POMDPs also requires preserving the temporal correlation in transition tuples. Thirdly, addressing the evaluation problem in parallel settings presents additional obstacles, such as correcting the cumulative bias in local updates.

Our proposed algorithm "Distributed Partially-observable Temporal Difference Learning (DPTD)", addresses all three challenges. For the first problem, we introduce linear function approximation for the value functions to simplify the structure of the estimated Q-function, converting the evaluation problem into a series of least-squares regression

$$\underset{\psi_h}{\text{minimize}} \ \frac{1}{2}\mathbb{E}^{\pi}\left[\left(\mathsf{Q}_h^{\pi} - \langle \phi_h(\bar{Z}_h), \psi_h \rangle\right)^2\right] \quad \text{for all } h \in \{0, 1, \ldots, H-1\} \tag{3}$$

ϕ_h provides a compact representation of POMDP transition dynamics, freeing us from computing a complex normalizing constant in the Bellman backup. To avoid exhaustive history searches, we adapt the Temporal-difference learning algorithm from MDPs to POMDPs (Cayci and Eryilmaz, 2024), which updates the Q-function by bootstrapping from its current estimate, allowing for direct optimization of ψ_h without acquiring full knowledge of Q_{h+1}^{π}. To tackle the second issue, we stack ψ_h across all time steps to form $\psi = (\psi_0^\top, \psi_1^\top, \ldots, \psi_{H-1}^\top)^\top \in \mathbb{R}^{Hd_Q}$. Then, we jointly optimize the H consecutive Bellman errors via $\underset{\psi}{\text{minimize}}\ \mathcal{E}^{\pi}(\psi)$, where

$$\mathcal{E}^{\pi}(\psi) := \frac{1}{2} \cdot \mathbb{E}\left[\sum_{h=0}^{H-1} \gamma^h \left(\mathsf{Q}_h^{\pi} - \langle \phi_h(\bar{Z}_h), \psi_h \rangle\right)^2\right] \tag{4}$$

This prevents compounding error and enables parallel updates of $\{\psi_h\}_{h=0}^{H-1}$, thereby speeding up optimization. We also sample entire roll-out trajectories $\bar{Z}_H$ from the POMDP model instead of individual transition tuples to preserve temporal correlation in data. Then we compute the gradient for ψ_h via [2]

$$\begin{aligned} g_h(\psi; \bar{Z}_{h+1}) := -\gamma^h \phi_h(\bar{Z}_h) \cdot \\ \left(r_h + \gamma \phi_{h+1}(\bar{Z}_{h+1})^\top \psi_{h+1} - \phi_h(\bar{Z}_h)^\top \psi_h\right) \end{aligned} \tag{5}$$

and stack g_h throughout the time horizon: $g(\psi; \bar{Z}_H) := \left[g_0(\psi; \bar{Z}_1)^\top, \ldots, g_{H-1}(\psi; \bar{Z}_H)^\top\right]^\top$, which are then employed to update the Q-function estimates at each time instant, all at once,

$$\psi^{r,k+1,m} \leftarrow \psi^{r,k,m} - \alpha_r \cdot g(\psi^{r,k,m}; \bar{Z}_H), \tag{6}$$

Here, α_r is the local learning rate in round r. We solve the third challenge by carefully designing the data parallel routine. We select a small local learning rate α_r to minimize the deviation of local parameters. Then we implement a weighted averaging strategy at the central server, where we compute $\widetilde{\psi}^{r+1}$ using the following relation: $\widetilde{\psi}^{r+1} := (1-\eta_r)\widetilde{\psi}^{r+1} + \eta_r \frac{1}{M}\sum_{m=1}^{M} \psi^{r,K,m}$. Since the gradients in the server possess less noise, it is safe to choose a larger global step size η_r for faster convergence. To deal with the inherent bias in TD learning brought by the bootstrapping technique (Sutton and Barto, 2018), we discard the inaccurate Q-function estimates from the initial iterations and use weight the parameters in the last $R - R_0$ rounds with quadratically increasing weights ω_r to obtain the final Q-function estimate: $\widehat{\psi} = \sum_{r=R_0}^{R} \omega_r \widetilde{\psi}^r$. Here, R_0 is a carefully chosen parameter. A pseudocode of DPTD can be found in Alg. 2.

[2] The term $g_h(\psi; \bar{Z}_{h+1})$ is the semi-gradient (Sutton and Barto, 2018).

Algorithm 2: DPTD

Input policy π, feature $\{\phi_h\}_{h=0}^{H-1}$, local l.r. $\{\alpha_r\}_{r=0}^{R-1}$, global l.r. $\{\eta_r\}_{r=0}^{R-1}$, weights $\{w_r\}_{r=0}^{R}$
Initialize $\psi^{0,0,m} \leftarrow 0$, $\widetilde{\psi}^r \leftarrow 0$.
for round $r = 0, 1, \ldots, R-1$ **do**
  //Distributed training
  for $m = 0, 1, \ldots, M-1$ in parallel **do**
    for local step $k = 0, 1, \ldots, K-1$ **do**
      Sample $\bar{Z}_H^{r,k,m}$ from π
      Compute $g(\psi^{r,k,m}; \bar{Z}_H^{r,k,m})$ via Eq. (5)
      $\psi^{r,k+1,m} \leftarrow \psi^{r,k,m} - \alpha_r g(\psi^{r,k,m}; \bar{Z}_H^{r,k,m})$
    end for
  end for
  Clients send $\psi^{r,K,m}$ to server
  //Centralized processing
  Averaging the estimates $\bar{\psi}^{r+1} \leftarrow \frac{1}{M}\sum_{m=0}^{M-1} \psi^{r,K,m}$
  Polyak averaging $\widetilde{\psi}^{r+1} \leftarrow (1-\eta_r)\,\widetilde{\psi}^r + \eta_r \bar{\psi}^{r+1}$
  Synchronize $\psi^{r+1,0,m} \leftarrow \widetilde{\psi}^{r+1}$
  if $r \geq R_0$: **then**
    Weighted sum $\widehat{\psi} = \widehat{\psi} + w_{r+1} \cdot \widetilde{\psi}^{r+1}$
  end if
end for
return $\{\widehat{Q}_h(\cdot) = \langle \phi_h(\cdot)\ ,\ \widehat{\psi}_h\ \rangle_{\mathbb{R}^{d_Q}}\}_{h=0}^{H-1}$

3.2 Policy Improvement

In this section, we describe our policy improvement routine, which we refer to as Distributed Partially-observable Natural Policy Gradient (DPPG). DPPG is a distributed, model-free algorithm for POMDPs inspired by the classic "Q-NPG" update for MDPs (Agarwal et al., 2019). In particular, DPPG updates the current policy parameter θ_r using the following update rule:

$$\theta_{r+1} \leftarrow \theta_r + \eta \cdot \omega,$$

where η is the step size and ω is a minimizer of the function $\mathcal{L}_{\mathsf{Q}}(\omega; \theta_r, \pi^{\theta_r})$, which is defined as

$$\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}} \sum_{h=0}^{H-1} \gamma^h \left(\omega^\top \nabla_\theta f_{\theta_r}(Z_h, A_h; h) - \mathsf{Q}_h^{\pi^{\theta_r}}(\bar{Z}_h)\right)^2 \tag{7}$$

We would like to point out that while this bears structural resemblance to the classic Q-NPG update rule (Agarwal et al., 2019), both the terms $f_{\theta_r}(Z_h, A_h; h)$ and $\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}$ exhibit a historical dependence due to the non-Markovian nature of the policy. We use parallelized linear regression (Haddadpour et al., 2019) to approximate the policy gradient ω_r, using samples of the function $\widehat{\mathcal{L}}_{\mathsf{Q}}(\omega; \theta; \bar{Z}_H) := \sum_{h=0}^{H-1} \gamma^h \left(\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h) - \omega^\top \nabla_\theta f_{\theta_r}(Z_h, A_h; h)\right)^2$, where $\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\cdot)$ denotes the estimate of $\mathsf{Q}_h^{\pi^{\theta_r}}(\cdot)$ obtained using the DPTD sub-routine. A pseudocode of DPPG is given in Alg. 3.

We also introduce two terms to evaluate the performance of solving the NPG direction ω. The first one is the statistical error, denoted as $\epsilon_{\text{stat}}(\omega; \theta_r) = \mathcal{L}_{\mathsf{Q}}(\omega; \theta_r; \pi^{\theta_r}) - \inf_\omega \mathcal{L}_{\mathsf{Q}}(\omega; \theta_r; \pi^{\theta_r})$, measures the accuracy of our linear regression solution. The second is the approximation error, $\epsilon_{\text{approx}}(\theta_r) = \inf_\omega \mathcal{L}_{\mathsf{Q}}(\omega; \theta_r; \pi^{\theta_r})$, represents the minimum error possible in approximating the Q-function with a linear combination of the coordinates of $\nabla_\theta f_\theta$. For a rich neural policy class, $\epsilon_{\text{approx}}(\theta_r)$ is very small (Wang et al., 2019). When we adopt inexact policy evaluation, we replace the Q-function with its empirical estimate $\widehat{Q}^{\pi^{\theta_r}}$ in these definitions.

Algorithm 3: DPPG

Input θ_0, evaluation procedure PolicyEval$(\cdot)$
for round r in $0, 1, \ldots, R-1$ **do**
  Evaluate policy in server $\mathsf{Q} \leftarrow$ PolicyEval$\left(\pi^{\theta_r}\right)$
  Send θ_r, Q to client agents.
  //Estimate gradient ω by distributed training
  for machine m in $1, 2, \ldots, M$ in parallel **do**
    Initialize $\omega_r^{0,m} \leftarrow 0$
    for local step k in $0, 1, \ldots, K-1$ **do**
      Collect trajectory $\bar{Z}_H^{k,m}$ by playing π^{θ_r}
      $\omega_r^{k+1,m} \leftarrow \omega_r^{k,m} - \zeta_k \nabla_\omega \widehat{L}_{\mathsf{Q}}\left(\omega_r^{k,m}; \theta_r, \bar{Z}_H^{k,m}\right)$
      //Aggregate ω in server
      if $k \equiv 0 \pmod{\mathrm{I}}$ **then**
        $\omega_r^{k+1} \leftarrow \frac{1}{M}\sum_{m=1}^{M} \omega_r^{k+1,m}$
        Synchronize $\omega_r^{k+1,m} \leftarrow \omega_r^{k+1}$
      end if
    end for
  end for
  //Update policy in server
  Update policy by $\theta_{r+1} \leftarrow \theta_r + \eta_r \cdot \omega_r$
end for
return Uniform mixture of $\theta_0, \theta_1, \ldots, \theta_{R-1}$

4 Performance Guarantees

Before we provide theoretical guarantees for Algorithms and 1, 2, and 3, we introduce several mild regularity conditions that are standard in the optimization literature (Bubeck et al., 2015), and we defer a discussion to Remarks 5 and 6.

Assumption 1 (Bounded Semi-gradient Norm). *$\exists G \in \mathbb{R}_+$, such that $\forall \psi: \quad \mathbb{E}_{Z_h} \ \|g(\psi; Z_h)\|_2 \leq G$*

Assumption 2 (Bounded Semi-gradient Noise). *There exists $\sigma_g^2 > 0$, such that $\mathrm{Var}\left[g_h(\psi; \bar{Z}_h)\right] \leq \sigma_g^2$*

Assumption 3 (Well-defined Features). *$\exists \ \mu_\phi \in \mathbb{R}_+$, s.t. $\mathbb{E}^{\pi^\theta}\left[\sum_{h=0}^{H-1} \gamma^h \ \phi_h(\bar{z}_h)\phi_h(\bar{z}_h)^\top\right] \succeq \mu_\phi \cdot \mathbb{I}_d$*

Assumption 4 (Bounded KL Divergence). *The optimal and the initial policy have a KL divergence below some finite D: $D_{KL}\left(\pi_h^\star(\cdot|z_h)||\pi_h^{\theta_0}(\cdot|z_h)\right) < D$*

Assumption 5 (Smooth Score Functions). *$\exists \beta > 0$, $\left|\nabla_\theta \ln \pi_h^{\theta'}(a_h|z_h) - \nabla_\theta \ln \pi_h^{\theta}(a_h|z_h)\right| \leq \beta \left\|\theta' - \theta\right\|_2$*

Assumption 6 (Bounded Gradient Noise). *There exists $\sigma_w^2 > 0$, s.t. $\mathbb{E}\left[\left\|\nabla_\omega \mathcal{L}_\mathcal{Q} - \nabla_\omega \ \widehat{\mathcal{L}}_\mathcal{Q}\right\|_2^2\right] \leq \sigma_w^2$*

Assumption 7 (Bounded Gradient Norm). *The optimal and estimated NPG gradients are bounded: $\exists W > 0$, $\left\|\omega_r^{k,m}\right\|_2$, $\|\omega^\star\|_2 \leq W$ a.s.*

Assumption 8 (Non-degenerate Fisher Matrix). *$\forall \theta$, the Fisher information matrix $\mathcal{F}_\theta$ of policy π^θ has positive eigenvalues greater than $\frac{\mu_F}{2}$ and less than $\frac{L}{2}$.*

Assumption 9 (Sample-path Coverage). *The concentratability coefficient between the trajectory probabilities induced by $\pi^\star$ and π^{θ_r} is almost surely finite: $\kappa_r := \mathbb{E}^{\pi^\star} \frac{d\mathbb{P}^{\pi^\star}}{d\mathbb{P}^{\pi^{\theta_r}}} < +\infty$. We also define $\kappa := \mathbb{E}_{\theta_{0:R-1}} \frac{1}{R}\sum_{r=0}^{R-1} \kappa_r$, where the expectation is taken w.r.t. the randomness in NPG.*

Under conditions 1 to 9, we obtain a theoretical guarantee for our Actor-Critic algorithm 1. The proofs can be found in Appendix 9.3.

Theorem 1. *There exists a learning rate design such that running Algorithm 1 with T_e steps of policy evaluation ensures functional suboptimality of*

$$\mathcal{V}^{\pi^\star} - \mathbb{E}_{\boldsymbol{\theta_1:R}}\left[\frac{1}{R}\sum_{r=1}^{R} \mathcal{V}^{\pi^{\boldsymbol{\theta_r}}}\right] \lesssim \underbrace{HW\sqrt{\frac{\beta D}{R}}}_{\text{Initialization}} + \underbrace{\frac{1}{\mu_F}\sqrt{\frac{\kappa H L \sigma_w^2}{MK}}}_{\text{Optimization}} + \underbrace{\frac{1}{\nu_\phi}\sqrt{\frac{\kappa H^2\left(d_Q + \sigma_g^2\right)}{MT_e}}}_{\text{Evaluation}} + \underbrace{\sqrt{\kappa H \epsilon_{\text{app}}}}_{\text{Approximation}} \tag{8}$$

Where we abbreviated $\frac{1}{R}\sum_{r=1}^{R} \mathbb{E}\ \epsilon_{\text{approx}}(\boldsymbol{\theta_r})$ as ϵ_{app}. Regularity constants σ_g^2, ν_ϕ, D, β, σ_w^2, L, and κ are defined in Assumptions 2, 3, 4, 5, 6, 8, and 9.

By carefully designing the data parallel routine, we can characterize the complexity of Algorithm 1 in the following corollary.

Corollary 1 (Complexity of Algorithm 1). *Pick $K = \frac{T^{1/2}}{M^{1/2}}$ and $T_e = K$, Algorithm 1 achieves ϵ-optimality after consuming $\widetilde{O}\left(\frac{C^2}{M\epsilon^4}\right)$ sample trajectories, where $C = H^2W^2\beta D + \frac{\kappa}{\mu_F^2}HL\sigma_g^2 + \kappa H^2\left(\frac{d_Q+\sigma_g^2}{{\nu_\phi}^2}\right)$. To achieve this, we need to send*

$$O\left((MT)^{1/2} + T^{5/6}M^{-1/6}\right)d_\theta + (MT)^{1/3}d_Q$$

floating-point parameters between the server and the clients. The total computation complexity is of order $O\left(TH(d_\theta + d_Q)\right)$.

Theorem 1 builds upon the analysis on the subroutines for policy evaluation and improvement, which we defer to Sections 4.1 and 4.2. In what follows, we discuss the implications of Theorem 1 and remark on several regularity constants.

Remark 2 (Complexity Trade-off). The computation complexity of Alg. 1 is irrelevant to the size of $\mathcal{O}$ and $\mathcal{S}$, because we replace an ideal planning oracle with TD learning and the NPG routine, both of which possesses an efficient implementation. Theorem 1 reveals how we trade the computation consumption off with extra error in optimizing ω_r with linear regression, evaluating the Q-function under linear function approximation, and an error floor that accounts for function approximation error inherent in NPG methods Agarwal et al. (2021). We can reduce the first three error terms in Eq. (8) with more sample consumption and policy updates, while the second and the third terms benefit from distributed training.

Remark 3 (Linear Speedup with Sublinear Communication). Lemma 1 shows that as parallel machines increase M times, the sample consumption per agent is reduced M-fold, showing a linear speedup in sample complexity. This explains why we can significantly accelerate POMDP solvers when thousands of agents are simulated in parallel (Rudin et al., 2022) in continuous settings. Meanwhile, our algorithm demands only a few sublinear communication rounds relative to the computation budget per agent T, showing an impressive communication efficiency.

Remark 4 (Comparison with MDP). Our complexity results do not go against the info-theoretical hardness of POMDPs (Papadimitriou and Tsitsiklis, 1987). Algorithm 1 only learns POMDPs whose value functions possess a linear structure, and the suboptimality suffers from an extra error floor due to approximation error. Optimizing POMDPs is still significantly harder than MDPs, which is revealed by additional dependency on $\sqrt{H}$ in the first term of Eq. (8) and a dependency on κ in the last three terms.

This can be attributed to history-dependent policies having a significantly more complex geometry than MDPs (Müller and Montúfar, 2021).

Next, we discuss several key regularity constants that appear in Theorem 1.

Remark 5 (Positive Definite Conditions). As for assumption 3, similar condition $\forall\theta$: $\mathbb{E}_{s\sim d^{\pi^\theta}}[\phi(s)\phi(s)^\top] \succeq \mu_\phi \cdot \mathbb{I}_d$ is commonly used in the theory of Actor-critic algorithms for MDPs (Bhandari et al., 2018; Qiu et al., 2021; Shen et al., 2023). We extend this to finite-horizon POMDPs by using multiple linear features ϕ_h to capture the temporal relations of POMDPs and substituting the stationary distribution d^{π^θ} with the sample path probability. Assumption 8 implies that $\mathcal{L}_\mathsf{Q}$ is μ_F-strongly convex, which ensures the uniqueness of $\omega^\star$. This is a common and not overly restrictive assumption in the theories of NPG algorithms for MDPs (Kakade, 2001; Liu et al., 2020; Yuan et al., 2023). A broad class of practical policies, such as Gaussian policies, satisfy this condition (see page 5 of (Liu et al., 2020)).

Remark 6 (Interpretation of Constant κ). In Assumption 9, κ_r equals $\mathbb{E}^{\pi^\star}\prod_{t=0}^{H-1}\frac{\pi_t^\star(A_t|Z_t)}{\pi_t^{\theta_r}(A_t|Z_t)}$. A small κ_r requires the learned policies to be exploratory enough to find the best actions. It generalizes similar restrictions from MDP theory (Yang et al., 2024), such as $\mathbb{E}_{(s,a)\sim d^{\pi^\star}}\left[\frac{d^{\pi^\star}(s,a)}{d^{\pi^{\theta_t}}(s,a)}\right] \leq C$, to POMDPs, however with the stationary distribution d replaced with sample path probability $\mathbb{P}$. The assumption is satisfied or relaxed when π belongs to tabular softmax or log-linear policies (Agarwal et al., 2021). Another work on POMDP theory (Cayci and Eryilmaz, 2024) also adopts similar assumptions to establish global convergence. For more analysis on κ, please refer to Appendix 9.1.2.

In the following two sections, we present the performance guarantee of Algorithm 2 and 3, along with a set of learning rate designs which are the building blocks of Theorem 1.

4.1 Policy Evaluation

The following lemma describes the complexities of Algorithm 2.

Lemma 1 (Complexity of Algorithm 2)**.** *Set* $\iota_r = \frac{a}{\mu_\phi(1-\gamma)(r+a)}$ *where* a *is any constant larger than* $\frac{9}{2}$. *Pick* $K = \min\left\{\frac{T^{2/3}}{M^{1/3}}, \sqrt{\frac{2}{e}\frac{H\sigma_g^2}{G}\frac{T}{M}}, \frac{T}{M}\right\}$. *For any local learning rate* $\alpha_r \leq \frac{\mu_\phi(1-\gamma)}{16T}$ *and global learning rate defined via* $\eta_r := \frac{\iota_r}{\alpha_r K}$, *Algorithm 2 returns an* ϵ*-accurate Q-function estimate using only*

$$N_{\text{eval}} = \widetilde{O}\left(\frac{1}{M\epsilon}\left(\frac{H\sigma_g^2}{{\nu_\phi}^2} + Hd_Q\right)\right) \tag{9}$$

sample trajectories per agent, within $R = \max\left\{(MT)^{1/3}, \frac{e}{2}M, \sqrt{\frac{e}{2}\frac{G}{H\sigma_g^2}MT}\right\}$ *rounds of communications. Moreover, given sufficient computation* $\frac{T}{M} \geq \frac{1}{{\nu_\phi}^2}\frac{G^2}{H\sigma_g^2}$, *we can pick* $K = \frac{T^{2/3}}{M^{1/3}}$ *and obtain* $\widetilde{O}\left(\frac{H\sigma_g^2+Hd}{{\nu_\phi}^2 MT}\right)$ *sample complexity using only* $R = (MT)^{1/3}$ *rounds of communications. The algorithm requires* $O(THd_Q)$ *operations per agent.*

Here, we denote $\nu_\phi = \mu_\phi(1-\gamma)$ as the condition number of TD learning. The proof of Lemma 1 is provided in Appendix 8.3.

Remark 7 (Interpretation of Lemma 1). Lemma 1 reveals that, when our optimization conditions are mild (ν_ϕ is large and the gradients are small), we can safely adopt longer local updates $K = \frac{T^{2/3}}{M^{1/3}}$ to reduce communication costs while allowing computation resources to drive the process of policy optimization. However, when the feature matrix becomes ill-defined (when μ_ϕ is small), we need to take fewer local steps to prevent the drift terms from blowing off. We derive these insights from a novel and precise characterization of the local drift terms in Lemma 5.

4.2 Policy Improvement

When the policy evaluation error is negligible, the NPG algorithm 3 itself possesses the following complexity bounds, whose proof is in Appendix 9.2.

Lemma 2 (Complexity of Algorithm 3)**.** *Pick* $K = \sqrt{\frac{T}{M}}$. *Set* $\zeta_k = \frac{4}{\mu_F(k+a)}$ *where* $a = qI+4$ *with* q *being a constant satisfying* $q\exp\left(-\frac{2}{q}\right) < \frac{L}{\mu_F}\sqrt{192\left(\frac{M+1}{M}\right)}$, $I = O\left(\frac{T^{1/3}}{M^{2/3}}\right)$, $\eta_r = \frac{\sqrt{2D/\beta}}{W\sqrt{R}}$. *When the function approximation error is small, Algorithm 3 promises* ϵ*-optimality with*

$$\widetilde{O}\left(\frac{H^4W^4\beta^2D^2 + \kappa^2/\mu_F^4 H^2L^2\sigma_w^2}{M\epsilon^4}\right)$$

sample trajectories for each parallel machine. In total, we need to communicate $O\left(M^{1/2}T^{1/2} + T^{5/6}M^{-1/6}\right)$ *times to synchronize the policy parameter* θ_r *with its gradients* ω_r. *The computation consumption is* $C_{\text{improve}} = O(THd_\theta)$.

Lemmas 1 and 2 reveal that both of our policy evaluation and policy improvement algorithms enjoy a lin-

ear speedup in sample complexity and sublinear communication complexity, which is the key to the performance guarantee of our Actor-critic framework.

5 Proof Outline

We outline the proofs for the main results in Section 4, and defer the details to Appendix 8 and 9.

To show Lemma 1, we introduce the potential function U to characterize the parameter's suboptimality.

Definition 1 (Potential Function in TD learning). $U(\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r+1}}) := \mathbb{E}\left\|\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r+1}} - \psi^\star\right\|_2^2$

The improvement of the potential function decomposes into two terms:

$$U(\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r+1}}) - U(\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}}) = \mathbb{E}\left\|\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r+1}} - \widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}}\right\|_2^2 + 2\mathbb{E}\left\langle \widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}} - \psi^\star \ , \ \widetilde{\boldsymbol{\psi}}^{\boldsymbol{r+1}} - \widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}}\right\rangle \tag{10}$$

We refer to the first term as the "Progression Distance" and the second as the "Progression Direction". They are governed by the following two lemmas, with their proofs available in Appendix 8.1.1 and 8.1.2.

Lemma 3 (Progression Direction). *For any constant $c \in \mathbb{R}_+$ the progression direction is controlled by*

$$\text{Progression Direction} \le (\eta_r \alpha_r K) \cdot \left[\frac{1}{2c} U\left(\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}}\right) + \mathbb{E}\left\langle \widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}} - \psi^\star, \mathbb{E}_{\boldsymbol{Z}_H} g\left(\widetilde{\boldsymbol{\psi}}^{\boldsymbol{r}}; \boldsymbol{Z}_H\right)\right\rangle + 8c \cdot \text{drift}(r)\right]$$

Lemma 4 (Progression Distance). *The progression distance has the following upper bound:*

$$\text{Progression Distance} \le (\eta_r \alpha_r K)^2 \cdot \left[4\left\|\mathbb{E}^{\pi}_{\overline{\boldsymbol{Z}}} g\left(\widetilde{\boldsymbol{\psi}}^{r}; \overline{\boldsymbol{Z}}\right)\right\|_2^2 + \frac{2H\sigma_g^2}{MK} + 64 \cdot \text{drift}(r)\right]$$

where drift(r) measures how local models deviate from the recent global update.

Definition 2 (Local Drift). *The local drift in round r is the average distance from the optimal parameter between consecutive local updates on each machine.* $drift(r) := \frac{1}{MK}\sum_{m,k} \mathbb{E}\left\|\boldsymbol{\psi}^{\boldsymbol{r,k,m}} - \bar{\boldsymbol{\psi}}^{\boldsymbol{r}}\right\|_2^2$

We control the drift term with a sharp bound derived from Riemannian integrals (see Appendix 8.1.3).

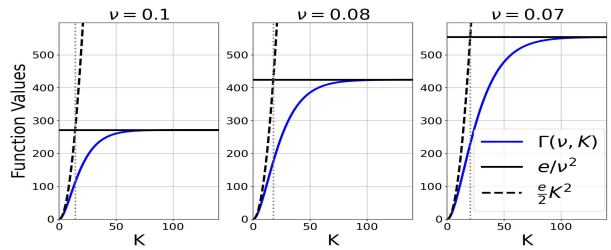
Lemma 5 (Local Drift Bound). *When α^r is less than $\frac{\mu_\phi(1-\gamma)}{16T}$, the drift term in the r-th round is governed by the following upper bound,*

$$\text{drift}(r) \le \Gamma(K;\nu_\phi)\left(H\sigma_g^2/K + 2G^2\right) \cdot \alpha_r^2$$

where $\Gamma(K;\nu_\phi) := \frac{e}{\nu_\phi{}^2}\left(1 - (1+\nu_\phi)e^{-\nu_\phi K}\right)$

We control $\Gamma(K;\nu_\phi)$ with different upper bounds in Lemma 13, which is illustrated in Fig. (2).

Figure 2: Local drift varies with K and ν_ϕ. When ν_ϕ is large, $\Gamma(K;\nu_\phi) \le \frac{e}{\nu_\phi{}^2}$, we can take more local steps to save communication. As $\nu_\phi \to 0$, the drift effect quickly deteriorates at a rate of $O(K^2)$, necessitating frequent synchronization.



Bringing Lemma 3, 4, 5 to Eq. (10), and sum up either sides with quadratically increasing weights after $r \ge R_0$, we obtain a bound on the Bellman error, which then implies Lemma 1.

Next, we extend the regret lemma in RL theory (Agarwal et al., 2021) to POMDPs, which links the functional suboptimality of output policies with the compatible function approximation error.

Lemma 6 (Regret Lemma for POMDPs).

$$V^{\pi^\star} - \mathbb{E}\frac{1}{R}\sum_{r=1}^{R} V^{\pi^{\boldsymbol{\theta_r}}} \le HW\sqrt{\frac{2\beta D}{R}} + \sqrt{\frac{\kappa H}{R}\sum_{r=1}^{R}\mathbb{E}L_{\widehat{\mathcal{Q}}}}$$

Here, $\widehat{\mathcal{Q}}$ is short for $\widehat{\mathcal{Q}}^{\pi^{\theta_r}}$ under inexact evaluation. Then, we decompose $L_{\widehat{\mathcal{Q}}}$ with simpler terms:

Lemma 7 (Error Decomposition).
$L_{\widehat{\mathcal{Q}}}(\omega;\theta,\pi^\theta) \le 2\epsilon_{\text{stat}}(\omega;\theta) + 2\epsilon_{\text{approx}}(\theta) + 4\mathcal{E}^{\pi^\theta}\left(\widehat{\psi}\right)$

Fortunately, statistical error enjoys a linear speedup,

Lemma 8 (Informal version of Lemma 16).
Under careful design, $\mathbb{E}\ \epsilon_{\text{stat}}(\boldsymbol{\omega};\boldsymbol{\theta_r}) \lesssim \frac{L}{\mu_F^2}\frac{\sigma_w^2}{MT}$

Theorem 1 directly result from Lemmas 6, 7 and 16.

6 Conclusion and Future Work

This work establishes the first theoretical framework for understanding the empirical success of distributed policy optimization in solving large POMDPs. The comprehensive complexity analysis, novel analytical techniques, and extensive explanations in this work are of independent interest to the community.

Building upon our findings, an exciting avenue is to analyze how exploration helps improve the convergence rates. We leave this to future work.

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The checklist follows the references. For each question, choose your answer from the three possible options: Yes, No, Not Applicable. You are encouraged to include a justification to your answer, either by referencing the appropriate section of your paper or providing a brief inline description (1-2 sentences). Please do not modify the questions. Note that the Checklist section does not count towards the page limit. Not including the checklist in the first submission won't result in desk rejection, although in such case we will ask you to upload it during the author response period and include it in camera ready (if accepted).

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   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [**Yes**/No/Not Applicable]
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/**Not Applicable**]
2. For any theoretical claim, check if you include:
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   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/**Not Applicable**]

Supplementary Material

7 Proof of Theorem 1

In theorem 1 we aim to show that

$$\mathcal{V}^{\pi^\star} - \mathbb{E}_{\boldsymbol{\theta_1:R}}\left[\frac{1}{R}\sum_{r=1}^{R}\mathcal{V}^{\pi^{\boldsymbol{\theta_r}}}\right] \lesssim \underbrace{HW\sqrt{\frac{\beta D}{R}}}_{\text{Initialization}} + \underbrace{\frac{1}{\mu_F}\sqrt{\frac{\kappa HL\sigma_w^2}{MK}}}_{\text{Optimization}} + \underbrace{\frac{1}{\nu_\phi}\sqrt{\frac{\kappa H^2\left(d_Q+\sigma_g^2\right)}{MT_e}}}_{\text{Evaluation}} + \underbrace{\sqrt{\kappa H\epsilon_{\text{app}}}}_{\text{Approximation}}$$

where $\epsilon_{\text{app}} := \frac{1}{R}\sum_{r=1}^{R}\mathbb{E}\ \epsilon_{approx}(\boldsymbol{\theta_r})$. This results is a consequence of the following several key lemmas.

First, we show a series of technical lemmas (Lemmas 3, 4 and 5) in Appendix 8.1 as a preparation.

Then, we prove Lemma 6 in Appendix 8.2.1 for softmax policy class:

$$\mathsf{V}^{\pi^\star} - \mathbb{E}\frac{1}{R}\sum_{r=0}^{R-1}\mathsf{V}^{\pi^{\theta_r}} \leq HW\sqrt{\frac{2\beta D}{R}} + \sqrt{\kappa H}\sqrt{\frac{1}{R}\sum_{r=0}^{R-1}\mathbb{E}\ \mathcal{L}_{\widehat{\mathsf{Q}}}\left(\omega_r;\theta_r,\pi^{\theta_r}\right)} \tag{11}$$

which controls the functional suboptimality with the initial error and compatible function approximation error. We then decompose the error $\mathcal{L}_{\widehat{\mathsf{Q}}}$ by Lemma 7, which is shown in Appendix 8.2.2:

$$L_{\widehat{\mathcal{Q}}}(\omega;\theta,\pi^\theta) \leq 2\epsilon_{\text{stat}}(\omega;\theta) + 2\epsilon_{\text{approx}}(\theta) + 4\mathcal{E}^{\pi^\theta}\left(\widehat{\boldsymbol{\psi}}\right) \tag{12}$$

The third step is to show that the Bellman error has an upper bound according to Lemma 1, whose proof is provided in Appendix 8.3:

$$\mathbb{E}\ \mathcal{E}^{\pi^\theta}\left(\widehat{\boldsymbol{\psi}}\right) \leq \frac{H(d_Q+\sigma_g^2)}{\nu_\phi^2 MT_e} \tag{13}$$

Finally, we invoke Lemma 16 to control the statistical error in Appendix 9.4:

$$\mathbb{E}\ \epsilon_{\text{stat}}(\boldsymbol{\omega;\theta_r}) \lesssim \frac{L}{\mu_F^2}\frac{\sigma_w^2}{MT} \tag{14}$$

We defer the proofs to the supporting lemmas to Appendix 8. Combining Eqs (11), (12), (13), and (14), we conclude the proof of Theorem 1.

8 Proof of Main Lemmas

In this section, we show the main lemmas in the proofs of Theorem 1.

8.1 Preparation

We first show three technical lemmas that will be used in the proof of Lemma 1.

8.1.1 Lemma 3

In Lemma 3, we aim to show that for any constant $c \in \mathbb{R}_+$:

$$\underbrace{2\cdot\mathbb{E}\left\langle\widetilde{\psi}^r-\psi^\star\ ,\ \widetilde{\psi}^{r+1}-\widetilde{\psi}^r\right\rangle}_{\text{Progression Direction}}$$
$$\leq\eta^r\alpha^rK\cdot\left[-2(1-\gamma)\ \underbrace{\mathbb{E}\ \mathcal{E}(\widetilde{\psi}^r)}_{\text{Bellman error}}+\frac{1}{c}\cdot\underbrace{\mathbb{E}\left\|\widetilde{\psi}^r-\psi^\star\right\|_2^2}_{\text{Potential function}}+16c\cdot\underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left\|\psi^{r,k,m}-\widetilde{\psi}^r\right\|_2^2}_{\text{Local drift}}\right]$$

Proof. We will heavily rely on the independence relation of successive rollout trajectories. For a precise characterization, we introduce a set of sigma algebras to describe the relationship between several random variables.

Definition 3. *Define the random events occurring before the derivation of $\psi_h^{r,k,m}$ as*

$$\mathcal{F}_h^{r,k,m}=\sigma\left(\psi^{r,k-1,m},\bar{z}_{h+1}^{r,k-1,m}\right)=\sigma\left(\bar{\psi}^r,\{\bar{z}_{h+1}^{r,t,m}\}_{t=0}^{k-1}\right)\qquad\mathcal{F}^{r,k,m}:=\bigcup_{h=0}^{H-1}\mathcal{F}_h^{r,k,m}$$

We further define the following sigma-algebras to characterize the randomness before we update other parameters:

$$\mathcal{F}_h^{r,k}:=\bigcup_{m=0}^{M-1}\mathcal{F}_h^{r,k,m}\qquad\mathcal{F}^{r,k}:=\bigcup_{h=0}^{H-1}\mathcal{F}_h^{r,k}\qquad\mathcal{F}_h^{r}:=\bigcup_{k=0}^{K}\mathcal{F}_h^{r,k}\qquad\mathcal{F}^{r}:=\bigcup_{h=0}^{H-1}\mathcal{F}_h^{r}$$

To prove Lemma 3, we unroll the update rule to exress the parameters by sample of the semi-gradients, before we telescope the sample gradients with their expectations and invoke their continuity conditions. First, we notice that from Algorithm 2,

$$\widetilde{\psi}^{r+1}=(1-\eta_r)\,\widetilde{\psi}^r+\eta_r\bar{\psi}^r=\widetilde{\psi}^r-\eta_r\alpha_r\cdot\frac{\sum_m}{M}\sum_{k=0}^{K-1}g_h(\psi^{r,k,m};\bar{z}_{h+1}^{r,k,m})\tag{15}$$

Eq. (15) suggests that the following relation holds for any realization of $\bar{\psi}^r,\bar{\psi}^{r+1}$ and $\psi^{r,k,m}$:

$$\begin{aligned}&2\left\langle\widetilde{\psi}^r-\psi^\star\ ,\ \widetilde{\psi}^{r+1}-\widetilde{\psi}^r\right\rangle\\=&-2\left(\eta^r\alpha^r\right)\sum_{h=0}^{H-1}\left\langle\widetilde{\psi}_h^r-\psi_h^\star,\frac{1}{M}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}g_h\left(\psi^{r,k,m};\bar{z}_{h+1}^{r,k,m}\right)-\mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}\ \ g_h\left(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m}\right)\right\rangle\\&+2\left(\eta^r\alpha^r\right)\frac{1}{M}\sum_{m=0}^{M-1}\sum_{h=0}^{H-1}\sum_{k=0}^{K-1}\left\langle\widetilde{\psi}_h^r-\psi_h^\star,\mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}\ \ g_h\left(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m}\right)\right\rangle\end{aligned}$$

By the towering rule, we see that the telescoped term has zero mean,

$$\begin{aligned}&\mathbb{E}\ \left\langle\widetilde{\psi}_h^r-\psi_h^\star\ ,\ g_h\left(\psi^{r,k,m};\bar{z}_{h+1}^{r,k,m}\right)-\mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}\ \ g_h\left(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m}\right)\right\rangle\\=&\mathbb{E}\ \left[\left\langle\widetilde{\psi}_h^r-\psi_h^\star,\mathbb{E}\ \left[g_h\left(\psi^{r,k,m};\bar{z}_{h+1}^{r,k,m}\right)-\mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}\ \ g_h\left(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m}\right)\right\rangle\left|\mathcal{F}^{r,k,m}\right.\right]\right]=0\end{aligned}$$

which implies that the left-hand-side in Lemma 3 equals

$$
\begin{aligned}
&2 \cdot \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \widetilde{\psi}^{r+1} - \widetilde{\psi}^r \right\rangle \\
=&2\left(\eta^r \alpha^r\right) \cdot \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \frac{1}{M}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1} \mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}} \ g\left(\psi^{r,k,m}; \bar{z}_{h+1}^{r,k,m}\right)\right\rangle \\
=&2\left(\eta^r \alpha^r\right) \cdot \sum_{k=0}^{K-1} \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \mathbb{E}_Z \ g\left(\widetilde{\psi}^r; Z\right)\right\rangle \\
&+ 2\left(\eta^r \alpha^r\right) \cdot \sum_{k=0}^{K-1} \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \frac{1}{M}\sum_{m=0}^{M-1} \mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}} \ g\left(\psi^{r,k,m}; \bar{z}_{h+1}^{r,k,m}\right) - \mathbb{E}_Z \ g\left(\widetilde{\psi}^r; Z\right)\right\rangle
\end{aligned}
$$

For the first term, Assumption 1 implies that

$$
\begin{aligned}
&2\left(\eta^r \alpha^r\right) \cdot \sum_{k=0}^{K-1} \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \mathbb{E}_Z \ g\left(\widetilde{\psi}^r; Z\right)\right\rangle \\
\leq& 2\eta^r \alpha^r K \cdot \left(\frac{\mathbb{E}\ \left\|\widetilde{\psi}^r - \psi^\star\right\|_2^2}{2} + \frac{\left\|\mathbb{E}_Z g\left(\widetilde{\psi}^r; Z\right)\right\|_2^2}{2}\right) \leq 2\eta^r \alpha^r K \cdot \left(\frac{\mathbb{E}\ \left\|\widetilde{\psi}^r - \psi^\star\right\|_2^2}{2} + \frac{G^2}{2}\right)
\end{aligned}
$$

For the second term, Lemma 10 implies

$$
\begin{aligned}
&2\left(\eta^r \alpha^r\right) \cdot \sum_{k=0}^{K-1} \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \frac{1}{M}\sum_{m=0}^{M-1} \mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}} \ g\left(\psi^{r,k,m}; \bar{z}_{h+1}^{r,k,m}\right) - \mathbb{E}_Z \ g\left(\widetilde{\psi}^r; Z\right)\right\rangle \\
\leq& 2\eta^r \alpha^r \sum_{k=0}^{K-1} \mathbb{E}\ \left(\frac{\left\|\widetilde{\psi}^r - \psi^\star\right\|}{2c} + \frac{c}{2}\frac{\sum_{m=0}^{M-1}}{M}\left\|\mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}} \ g\left(\psi^{r,k,m}; \bar{z}_{h+1}^{r,k,m}\right) - \mathbb{E}_Z \ g\left(\widetilde{\psi}^r; Z\right)\right\|_2^2\right) \\
\leq& 2\eta^r \alpha^r \sum_{k=0}^{K-1} \mathbb{E}\ \left(\frac{\left\|\widetilde{\psi} - \psi^\star\right\|_2^2}{2c} + \frac{c}{2}\frac{\sum_{m=0}^{M-1}}{M} 16\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right)
\end{aligned}
$$

which results in

$$
\underbrace{2 \cdot \mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star \ , \ \widetilde{\psi}^{r+1} - \widetilde{\psi}^r \right\rangle}_{\text{Progression Direction}} \leq \eta^r \alpha^r K \left[\left(1 + \frac{1}{c}\right) \mathbb{E}\ \left\|\widetilde{\psi}^r - \psi^\star\right\|_2^2 + G^2 + 16c \cdot \underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1} \mathbb{E}\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2}_{\text{Local drift}}\right]
$$

□

Before we proceed to prove the next lemma (Lemma 4), we need to characterize how the variance in sample gradients is reduced by sampling independent rollout trajectories from the same policy.

Lemma 9 (Variance Reduction)**.**

$$
\mathbb{E}\left\|\frac{1}{MK}\sum_{m=0}^{M}\sum_{k=0}^{K-1} g_h(\psi^{r,k,m}; \bar{z}_{h+1}^{r,k,m}) - \mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}^{\pi} \ g_h(\psi^{r,k,m}; \bar{Z}_{h+1}^{r,k,m})\right\|_2^2 \leq \frac{1}{MK}\sigma_g^2
$$

This result is a smooth generalization of Lemma 7 in (Wang et al., 2023) and we omit the proof for brevity.

We also need to describe the continuity of the semi-gradients, which is summarized in the following lemma.

Lemma 10 (Lipchitzness of the Expected Semi-gradient)**.** *The ensemble mean of the stacked semi-gradient has Lipschitz constant* 4*:* $\left\|\mathbb{E}_{\bar{Z}_{h+1}}\ g(\psi^1;\bar{Z}_{h+1})-\mathbb{E}_{\bar{Z}_{h+1}}\ g(\psi^2;\bar{Z}_{h+1})\right\|_2 \leq 4\left\|\psi^1-\psi^2\right\|_2$

Proof. Under linear representation, the ensemble mean of the semi-gradient at each step takes the following form:

$$\begin{aligned}
&\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^1;\bar{Z}_{h+1})\\
=&-\gamma^h\mathbb{E}\ \left[\phi_h(\bar{Z}_H)(r(\bar{Z}_H)+\gamma\phi_{h+1}(\bar{Z}_{h+1})^\top\psi^1_{h+1}-\phi_h(\bar{Z}_H)^\top\psi^1_h)\right]\\
=&-\gamma^h\mathbb{E}\ \left[\phi_h(\bar{Z}_H)r(\bar{Z}_H)\right]-\gamma^{h+1}\mathbb{E}\left[\phi_h(\bar{Z}_H)\phi_{h+1}(\bar{Z}_{h+1})\right]^\top\psi^1_{h+1}-\gamma^h\mathbb{E}\phi_h(\bar{Z}_H)\phi_h(\bar{Z}_H)^\top\psi^1_h
\end{aligned}$$

Triangle inequality and Cauchy-Schwartz inequality suggest that

$$\begin{aligned}
&\left\|\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^1;\bar{Z}_{h+1})-\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^2;\bar{Z}_{h+1})\right\|_2\\
=&\left\|-\gamma^{h+1}\mathbb{E}\left[\phi_h(\bar{Z}_H)\phi_{h+1}(\bar{Z}_{h+1})\right]^\top\left(\psi^1_{h+1}-\psi^2_{h+1}\right)-\gamma^h\mathbb{E}\phi_h(\bar{Z}_H)\phi_h(\bar{Z}_H)^\top\left(\psi^1_h-\psi^2_h\right)\right\|_2\\
=&\gamma^{h+1}\left\|\mathbb{E}\left[\phi_h(\bar{Z}_H)\phi_{h+1}(\bar{Z}_{h+1})\right]^\top\left(\psi^1_{h+1}-\psi^2_{h+1}\right)\right\|_2+\gamma^h\left\|\mathbb{E}\phi_h(\bar{Z}_H)\phi_h(\bar{Z}_H)^\top\left(\psi^1_h-\psi^2_h\right)\right\|_2\\
=&\gamma^{h+1}\left\|\psi^1_{h+1}-\psi^2_{h+1}\right\|_2+\gamma^h\left\|\psi^1_h-\psi^2_h\right\|_2
\end{aligned}$$

which implies

$$\left\|\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^1;\bar{Z}_{h+1})-\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^2;\bar{Z}_{h+1})\right\|_2^2\leq 2\gamma^{2h+2}\left\|\psi^1_{h+1}-\psi^2_{h+1}\right\|_2^2+2\gamma^{2h}\left\|\psi^1_h-\psi^2_h\right\|_2^2$$

Stacking semi-gradients according to Eq. (6), we conclude

$$\begin{aligned}
&\left\|\mathbb{E}_{\bar{Z}_H}\ g(\psi^1;\bar{Z}_H)-\mathbb{E}_{\bar{Z}_{h+1}}\ g(\psi^2;\bar{Z}_{h+1})\right\|_2^2=\sum_{h=0}^{H-1}\left\|\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^1;\bar{Z}_{h+1})-\mathbb{E}_{\bar{Z}_{h+1}}\ g_h(\psi^2;\bar{Z}_{h+1})\right\|_2^2\\
\leq&\sum_{h=0}^{H-1}2\gamma^{2h+2}\left\|\psi^1_{h+1}-\psi^2_{h+1}\right\|_2^2+2\gamma^{2h}\left\|\psi^1_h-\psi^2_h\right\|_2^2\leq 4\sum_{h=0}^{H-1}\left\|\psi^1_h-\psi^2_h\right\|_2^2=4\left\|\psi^1-\psi^2\right\|_2^2
\end{aligned}$$

which is what we desired. □

Building upon Lemma 9 and 10, we are ready to show Lemma 4.

8.1.2 Proof of Lemma 4

In Lemma 4 we claim that

$$\underbrace{\mathbb{E}\ \left\|\widetilde{\psi}^{r+1}-\widetilde{\psi}^r\right\|_2^2}_{\text{Progression Distance}}\leq(\eta^r\alpha^rK)^2\cdot\left[4\left\|\mathbb{E}^\pi_{\bar{Z}}g\left(\widetilde{\psi}^r;\bar{Z}\right)\right\|_2^2+64\underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m}-\widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}}+2\underbrace{\frac{H\sigma_g^2}{MK}}_{\text{Variance}}\right]$$

Proof. Similar to the proofs in Appendix 8.1.1, the update rule in Eq. (15) and Young's inequality suggests

that the progression distance decomposes into two terms:

$$\begin{aligned}\mathbb{E}\left\|\widetilde{\psi}^{r+1}-\widetilde{\psi}^{r}\right\|_2^2 &= (\eta^r\alpha^r)^2\cdot\sum_{h=0}^{H-1}\mathbb{E}\left\|\frac{1}{M}\sum_{m=0}^{M}\sum_{k=0}^{K-1}g_h(\psi^{r,k,m};\bar{z}_{h+1}^{r,k,m})\right\|_2^2\\ &\le \underbrace{2\,(\eta^r\alpha^r)^2\sum_{h=0}^{H-1}\mathbb{E}\left\|\frac{1}{M}\sum_{m=0}^{M}\sum_{k=0}^{K-1}g_h(\psi^{r,k,m};\bar{z}_{h+1}^{r,k,m})-\mathbb{E}^{\pi}_{\bar{Z}_{h+1}^{r,k,m}}\,g_h(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m})\right\|_2^2}_{:=A_1}\\ &\quad+\underbrace{2\,(\eta^r\alpha^r)^2\sum_{h=0}^{H-1}\mathbb{E}\left\|\frac{1}{M}\sum_{m=0}^{M}\sum_{k=0}^{K-1}\mathbb{E}^{\pi}_{\bar{Z}_{h+1}^{r,k,m}}\,g_h(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m})\right\|_2^2}_{:=A_2}\end{aligned}\tag{16}$$

The first term, A_1, is associated with the variance of the noise in the sampling process, while the second term, A_2, is related to the local drift defined in 2 and the bias in the semi-gradient.

Lemma 9 indicates that the variance term is governed by the following bound

$$A_1 \le 2\,(\eta^r\alpha^r)^2\sum_{h=0}^{H-1}\frac{K\sigma_g^2}{M} = 2(\eta^r\alpha^r)^2K\frac{H\sigma_g^2}{M}\tag{17}$$

The bias term is related to the Bellman error. Specifically, Jensen's inequality suggests that

$$\begin{aligned}&\sum_{h=0}^{H-1}\mathbb{E}\left\|\frac{1}{M}\sum_{m=0}^{M}\sum_{k=0}^{K-1}\mathbb{E}^{\pi}_{\bar{Z}_{h+1}^{r,k,m}}\,g_h(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m})\right\|_2^2\\ \le&\sum_{h=0}^{H-1}K^2\cdot\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\mathbb{E}\left[\left\|\mathbb{E}^{\pi}_{\bar{Z}_{h+1}^{r,k,m}}\,g_h(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m})-\mathbb{E}^{\pi}_{\bar{Z}}\,g_h(\widetilde{\psi}^{r};\bar{Z})+\mathbb{E}^{\pi}_{\bar{Z}}\,g_h(\widetilde{\psi}^{r};\bar{Z})\right\|_2^2\middle|\mathcal{F}^{r,k,m}\right]\right]\end{aligned}$$

With a telescoping trick, we observe that

$$\begin{aligned}&\sum_{h=0}^{H-1}\mathbb{E}\left\|\frac{1}{M}\sum_{m=0}^{M}\sum_{k=0}^{K-1}\mathbb{E}^{\pi}_{\bar{Z}_{h+1}^{r,k,m}}\,g_h(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m})\right\|_2^2\\ \le&2K^2\cdot\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\left(\mathbb{E}\left[\mathbb{E}\left[\left\|\mathbb{E}^{\pi}_{\bar{Z}_{h+1}^{r,k,m}}\,g(\psi^{r,k,m};\bar{Z}_{h+1}^{r,k,m})-\mathbb{E}^{\pi}_{\bar{Z}}\,g(\widetilde{\psi}^{r};\bar{Z})\right\|_2^2\right]\middle|\mathcal{F}^{r,k,m}\right]+\sum_{h=0}^{H-1}\left\|\mathbb{E}^{\pi}g_h(\widetilde{\psi}^{r};\bar{Z})\right\|_2^2\right)\\ \le&32K^2\cdot\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m}-\widetilde{\psi}^{r}\right\|_2^2\right]+2K^2\sum_{h=0}^{H-1}\left\|\mathbb{E}^{\pi}g_h(\widetilde{\psi}^{r};\bar{Z})\right\|_2^2\end{aligned}\tag{18}$$

where the first step is by Young's inequality and the second is due to the Lipchitzness of the average semi-gradients (Lemma 10). The proof is completed after we bring Eq. (18), (17) to Eq. (16). □

In what follows, we will show how the upper bound on the local-drift term scales with the gradient variance and the learning rate (Lemma 5). As a preparation work, we need to extend the definition of Bellman error, and relate the semi-gradients with the Bellman error.

To allow for more flexibility, we extend the definition of Bellman error in Eq. (3) to any two stacked parameters:

Definition 4 (Extended Bellman Error)**.** *Given two parameters ψ^1 and ψ^2. Let $\widehat{\mathsf{Q}}^i$ denote the Q-function estimate constructed from $\widehat{\mathsf{Q}}_h^i = \langle\phi_h(\bar{Z}_h),\psi_h^i\rangle$ for $i\in\{1,2\}$. Define*

$$\mathcal{E}^{\pi}(\psi_1,\psi_2) := \mathbb{E}^{\pi}\left[\sum_{h=0}^{H-1}\gamma^h\frac{\left(\widehat{\mathsf{Q}}_h^1(\bar{Z}_h)-\widehat{\mathsf{Q}}_h^2(\bar{Z}_H)\right)^2}{2}\right] = (\psi_1-\psi_2)^{\top}\,\mathbb{E}\left[\sum_{h=0}^{H-1}\gamma^h\;\phi_h(\bar{z}_h)^{\top}\phi_h(\bar{z}_h)\right](\psi_1-\psi_2)$$

It is straightforward to see that $\mathcal{E}^\pi(\psi, \psi^\star) = \mathcal{E}^\pi(\psi)$. Moreover, condition 3 ensure that

$$\|\psi - \psi^\star\|_2^2 \leq \frac{1}{\mu_\phi}\ \mathcal{E}(\psi), \quad \|\psi_1 - \psi_2\|_2^2 \leq \frac{1}{\mu_\phi}\ \mathcal{E}^\pi(\psi_1, \psi_2) \tag{19}$$

Comparing Eq. (49) with Eq.(19), we discover the relationship between the potential function and the Bellman error:

$$\mu_\phi \cdot U\left(\widetilde{\psi}^r\right) \leq \mathbb{E}\ \mathcal{E}\left(\widetilde{\psi}^r\right) \leq U\left(\widetilde{\psi}^r\right) \tag{20}$$

Now we are ready to state the following lemma.

Lemma 11 (First-order Relation for Extended Bellman Error)**.**

$$\mathbb{E}\left\langle \psi^1 - \psi^2\ ,\ \mathbb{E}_{\bar{Z}}\left[g(\psi^1; \bar{Z})\right] - \mathbb{E}_{\bar{Z}}\left[g(\psi^2; \bar{Z})\right]\right\rangle \leq -(1-\gamma)\mathbb{E}\ \mathcal{E}(\psi^1; \psi^2)$$

Proof.

$$g_h(\psi^1; \bar{z}) = \gamma^h \phi_h\left(r + \gamma \phi_{h+1}^\top \psi_{h+1}^1 - \psi_h^\top \psi_h^1\right) g_h(\psi^2; \bar{z}) = \quad \gamma^h \phi_h\left(r + \gamma \phi_{h+1}^\top \psi_{h+1}^2 - \psi_h^\top \psi_h^2\right)$$

So we have

$$\begin{aligned}
LHS =& \mathbb{E}\sum_{h=0}^{H-1}\langle \psi_h^1 - \psi_h^2, g_h(\psi_1) - g_h(\psi_2)\rangle = -\mathbb{E}\sum_{h=0}^{H-1}\langle \psi_h^1 - \psi_h^2, -\gamma^h \phi_h(\gamma \phi_{h+1}^\top(\psi_{h+1}^1 - \psi_{h+1}^2)) - \phi_h^\top(\psi_h^1 - \psi_h^2)\rangle \\
=& -\mathbb{E}\sum_{h=0}^{H-1}\gamma^{h+1}\langle \phi_h, \psi_h^1 - \psi_h^2\rangle \cdot \langle \phi_{h+1}, \psi_{h+1}^1 - \psi_{h+1}^2\rangle + \gamma^h \langle \phi_h, \psi_h^1 - \psi_h^2\rangle \\
=& \mathbb{E} - \gamma \sum_{h=0}^{H-1}\gamma^h u_h u_{h+1} + 2\mathcal{E}\left(\psi_1, \psi_2\right) \leq \gamma \mathbb{E}\sum_{h=0}^{H-1}\gamma^h\left(\frac{u_h^2}{2\gamma} + \frac{\gamma u_h^2}{2}\right) - 2\mathcal{E}\left(\psi_1, \psi_2\right) \\
=& -\mathcal{E}\left(\psi^1, \psi^2\right) + \gamma(\mathcal{E}\left(\psi^1, \psi^2\right) - \frac{u_0^2}{2}) + 2\mathcal{E}\left(\psi_1, \psi_2\right) \leq -(1-\gamma)\mathcal{E}\left(\psi_1, \psi_2\right) = RHS
\end{aligned}$$

We remark that we write $\phi_t(\bar{z}_t)^\top(\psi_t^k - \psi^\star)$ as u_t for brevity. □

Next, we prove Lemma 5.

8.1.3 Proof of Lemma 5

We would like to show that

$$\underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}} \leq (\alpha^r)^2 \cdot \Gamma(K; \nu_\phi) \cdot \left(\frac{H\sigma_g^2}{K} + 2G^2\right)$$

where

$$\Gamma(K; \nu_\phi) := e \cdot \frac{1 - (1 + \nu_\phi K)e^{-\nu_\phi K}}{{\nu_\phi}^2} \tag{21}$$

$\Gamma(K; \nu_\phi)$ approaches $\frac{eK^2}{2}$ when $\gamma = 1$ or $\mu_\phi = 0$, while is dominated by $\frac{e}{{\mu_\phi}^2(1-\gamma)^2}$ when $0 < \alpha^r < \frac{\mu_\phi(1-\gamma)}{16}$. This result directly implies that,

- When $\mu_\phi \neq 0$ and $\gamma \neq 1$ and the learning rate satisfies $\alpha^r < \frac{\mu_\phi(1-\gamma)}{16}$,

$$\underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}} \leq \frac{e \cdot \alpha_r^2}{(1-\gamma)^2{\mu_\phi}^2}\left(\frac{H\sigma_g^2}{K} + 2G^2\right)$$

- When $\mu_\phi = 0$ or $\gamma = 1$ and the learning rate satisfies $\alpha^r < \frac{\mu_\phi(1-\gamma)}{16}$,

$$\underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m}-\widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}} \leq \frac{e\cdot(\alpha^r)^2K^2}{2}\left(\frac{H\sigma_g^2}{K}+2G^2\right)$$

which is the cornerstone of Lemma 13.

Proof. The proof is a stronger version of proof of Lemma 9 in (Wang et al., 2023). We express the parameter ψ in terms of the sampled semi-gradients, then we use a telescoping trick to replace the sampled gradients with their expected value.

$$\begin{aligned}
&\underbrace{\mathbb{E}\left[\left\|\psi^{r,k,m}-\widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}}\\
=&\sum_h \mathbb{E}\left[\left\|\psi_h^{r,k-1,m}-\widetilde{\psi}_h^r+\psi_h^{r,k,m}-\psi_h^{r,k-1,m}\right\|_2^2\right]\\
=&\sum_h \mathbb{E}\left[\left\|\psi_h^{r,k-1,m}-\widetilde{\psi}_h^r+\alpha^r g_h(\psi^{r,k-1,m};\bar{z}_{h+1}^{r,k-1,m})\right\|_2^2\right]\\
=&\sum_h \mathbb{E}\left[\left\|\psi_h^{r,k-1,m}-\widetilde{\psi}_h^r+\alpha^r \mathbb{E}_{\bar{Z}_{h+1}^{r,k-1,m}}\ g_h(\psi^{r,k-1,m};\bar{z}_{h+1}^{r,k-1,m})\right\|_2^2\right]\\
&+(\alpha^r)^2\cdot\mathbb{E}\ \left[\left\|g_h(\psi^{r,k-1,m};\bar{z}_{h+1}^{r,k-1,m})-\mathbb{E}_{\bar{Z}_{h+1}^{r,k-1,m}}\ g_h(\psi^{r,k-1,m};\bar{z}_{h+1}^{r,k-1,m})\right\|_2^2\right]\\
\leq&\sum_h \mathbb{E}\left[\left\|\psi_h^{r,k-1,m}-\widetilde{\psi}_h^r+\alpha^r \mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}\ g_h(\psi^{r,k-1,m};\bar{z}_{h+1}^{r,k,m})\right\|_2^2\right]+(\alpha^r)^2\cdot H\sigma_g^2
\end{aligned}$$

The third step is due to the cross term being zero if we condition on $\mathcal{F}^{r,k-1,m}$. We proceed by telescope again to replace the gradients at each parameter $\psi^{r,k,m}$ with that at the last synchronization $\widetilde{\psi}^r$.

$$\begin{aligned}
&\underbrace{\mathbb{E}\left[\left\|\psi^{r,k,m}-\widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}}\\
\leq&(\alpha^r)^2\cdot H\sigma_g^2+(\alpha^r)^2\left(1+\frac{1}{\xi}\right)\left\|\mathbb{E}_{Z_{h+1}}\ g(\widetilde{\psi}^r;Z_{h+1})\right\|_2^2\\
&+(1+\xi)\sum_h\mathbb{E}\left[\|\|\underbrace{\psi_h^{r,k-1,m}-\widetilde{\psi}_h^r}_{\text{I}}+\underbrace{\alpha^r\mathbb{E}_{\bar{Z}_{h+1}^{r,k,m}}\ g_h(\psi^{r,k-1,m};\bar{z}_{h+1}^{r,k,m})-\alpha^r\mathbb{E}_{Z_{h+1}}\ g_h(\widetilde{\psi}^r;Z_{h+1})}_{\text{II}}\|\|_2^2\right]
\end{aligned}$$

This relation holds for any positive ξ due to Young's inequality and later we will select an appropriate value of ξ that balances the terms when we unroll k to 0. Next, we associate term II with term I by the 4-Lipschitzness of the semi-gradient stated in Lemma 10:

$$\left\|\mathbb{E}_{\bar{Z}_{h+1}}\ g(\psi^1;\bar{Z}_{h+1})-\mathbb{E}_{\bar{Z}_{h+1}}\ g(\psi^2;\bar{Z}_{h+1})\right\|_2\leq 4\left\|\psi^1-\psi^2\right\|_2$$

The cross term between I and II can be derived from Eq. (19) and Lemma 11.

$$\mathbb{E}\left\langle\psi^1-\psi^2\ ,\ \mathbb{E}_{\bar{Z}}\left[g(\psi^1;\bar{Z})\right]-\mathbb{E}_{\bar{Z}}\left[g(\psi^2;\bar{Z})\right]\right\rangle\leq-(1-\gamma)\mathbb{E}\ \mathcal{E}(\psi^1;\psi^2)\leq-(1-\gamma)\mu_\phi\cdot\mathbb{E}\left\|\psi^1-\psi^2\right\|_2^2$$

provided that the feature covariate matrix is positive-definite. By rearranging terms and then taking the ensemble mean, we see that the drift term subjects to a linear dynamic equation:

$$\text{drift}(k;r)\leq A\cdot\text{drift}(k;r)+B \tag{22}$$

where

$$\begin{aligned}\text{drift}(k;r) :=& \frac{\sum_m}{M}\mathbb{E}\left[\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right] \\ A :=& (1+\xi)\left(1+16(\alpha^r)^2 - 2\alpha^r(1-\gamma)\mu_\phi\right) \\ B :=& (\alpha^r)^2 \cdot \left(H\sigma_g^2 + \left(1+\frac{1}{\zeta}\right)G^2\right)\end{aligned} \tag{23}$$

The synchronization rule (line 16 in Algorithm 2) suggests that $\text{drift}(0;r) = 0$, which then implies

$$\text{drift}(k;r) \leq A^k \text{drift}(0;r) + B \cdot \sum_{t=0}^{k} A^t = A^k \cdot 0 + B \cdot \sum_{t=0}^{k} A^t \leq kA^k \cdot B$$

It is also convenient to abbreviate the local drift as $\text{drift}(r)$. and the definitions of the drift terms naturally suggests that

$$\text{drift}(r) = \underbrace{\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}} = \frac{1}{K}\sum_{k=0}^{K-1}\underbrace{\text{drift}(k;r)}_{\text{Ensemble drift}} < B \cdot \frac{1}{K}\sum_{k=0}^{K-1} k \cdot A^k \tag{24}$$

We will compute the right-hand-side of Eq. (24) and then complete the proofs.

First we would like to simplify the scaling term A in Eq.(22) and sure that unrolling k will not cause any terms to diverge when K is large. Pick $\xi = \frac{1}{K}$ and $\alpha^r < \frac{\mu_\phi(1-\gamma)}{16}$, we can guarantee that $(1+\xi)^K < e$ and that $\left(1+16(\alpha^r)^2 - 2\alpha^r(1-\gamma)\mu_\phi\right) \leq 1-(1-\gamma)\mu_\phi\alpha^r \leq e^{-(1-\gamma)\mu_\phi\alpha^r}$ Next we connect the summation to a Riemannian integral and provide sharp bounds for all $K \in \mathbb{Z}_+$:

$$\begin{aligned}\text{drift}(r) <& eB \cdot \frac{1}{K}\int_0^K xe^{-ax}dx\Big|_{a=(1-\gamma)\mu_\phi} = eB \cdot \frac{1-(1+aK)\cdot e^{-aK}}{a^2K}\Big|_{a=(1-\gamma)\mu_\phi} \\ =& \Gamma(\gamma,\mu_\phi,K)\cdot\frac{B}{K} \leq \begin{cases}\frac{eB}{(1-\gamma)^2{\mu_\phi}^2K} & a>0 \\ \frac{eBK}{2} & a\to 0^+\end{cases}\end{aligned}$$

For reader's reference, we plot the summation with its upper bounds evaluated at $\mu_\phi(1-\gamma) = 0.07$: which

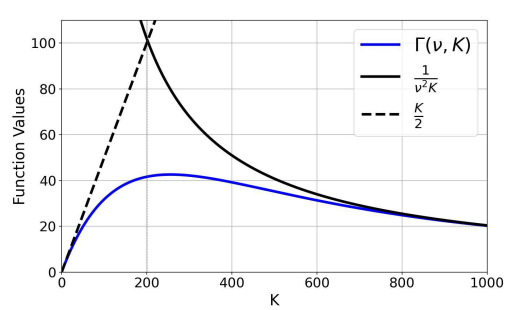

Figure 3: Upper bounds on drift(r) across different K. Curves evaluated at $B = 1, \mu_\phi(1-\gamma) = 0.007$

clearly shows that the bound $\frac{eB}{(1-\gamma)^2{\mu_\phi}^2K}$ is approximately effective when K is large, while $\frac{eBK}{2}$ dominates the summation when K or $\mu_\phi(1-\gamma)$ is small.

We complete the proof by substituting B with its definition in Eq. (23), where we select $\xi = \frac{1}{K}$. We will also need to use the simple fact that $\left(1+\frac{1}{K}\right) < 2$. □

8.2 Error Decomposition

8.2.1 Proof of Lemma 6

In this lemma we aim to show that

$$\mathsf{V}^{\pi^\star} - \mathbb{E}\frac{1}{R}\sum_{r=0}^{R-1}\mathsf{V}^{\pi^{\theta_r}} \leq HW\sqrt{\frac{2\beta D}{R}} + \sqrt{\kappa H}\sqrt{\frac{1}{R}\sum_{r=0}^{R-1}\mathbb{E}\ \mathcal{L}_{\widehat{\mathsf{Q}}}\left(\omega_r;\theta_r,\pi^{\theta_r}\right)}$$

The proof is divided into two steps.

Step 1: We will first show that

$$\mathsf{V}^{\pi^\star} - \mathbb{E}\frac{1}{R}\sum_{r=0}^{R-1}\mathsf{V}^{\pi^{\theta_r}} \leq HW\sqrt{\frac{2\beta D}{R}} + \sqrt{\kappa H}\sqrt{\frac{1}{R}\sum_{r=0}^{R-1}\mathbb{E}\ \mathcal{L}_{\widehat{\mathsf{A}}}\left(\omega_r;\theta_r,\pi^{\theta_r}\right)}$$

Here, κ is a regularity constant introduced in Eq. (9), $\widehat{A}$ is defined via $\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}(\bar{Z}_h) = \widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h) - \mathbb{E}_{A_h\sim\pi^\theta(\cdot|Z_h)}\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(Z_h,A_h)$, and $\mathcal{L}_{\widehat{\mathsf{A}}}$ is the compatible function approximation error, defined as

$$\mathcal{L}_{\widehat{\mathsf{A}}}(\omega;\theta_r,\pi^{\theta_r}) := \mathbb{E}^\pi\left[\sum_{h=0}^{H-1}\gamma^h\left(\omega^\top\nabla_\theta\ln\pi_h^\theta(A_h|Z_H) - \widehat{\mathsf{A}}_h^{\pi^{\theta_r}}(Z_H,A_h)\right)^2\right] \tag{25}$$

Step 2: Then, we prove that for softmax policies, $\mathcal{L}_{\widehat{\mathsf{A}}}(\omega;\theta_r,\pi^{\theta_r}) \leq \mathcal{L}_{\widehat{\mathsf{Q}}}(\omega;\theta_r,\pi^{\theta_r})$.

In what follows we show the two steps separately.

Proof of Step 1. The following proof is a smooth generalization of NPG regret bounds to POMDPs, which is inspired by (Agarwal et al., 2021; Cayci and Eryilmaz, 2024). Note that our proof adopts a different definition for κ compared with (Cayci and Eryilmaz, 2024), and our policy class is no longer limited to Markov policies like (Agarwal et al., 2021). We would like to introduce some functions for the convenience of our analysis. First, let us define

$$\epsilon^\star(\omega_r;\theta_r) := \mathbb{E}^{\pi^\star}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right) - \omega_r^\top\nabla_\theta\ln\pi_h^{\theta_r}(A_h|Z_h)\right) \tag{26}$$

We also define the following potential function under assumptions 5 and 9,

$$U(\pi^{\theta_r};\pi^\star) := \mathbb{E}^{\pi^\star}\sum_{h=0}^{H-1}\gamma^h D_{\mathrm{KL}}\left(\pi_h^\star\left(\cdot\mid Z_h\right)\|\pi_h^r\left(\cdot\mid Z_h\right)\right) < +\infty \tag{27}$$

which will be used as a tool to link the functional suboptimality $\mathsf{V}^{\pi^\star} - \mathsf{V}^{\pi^{\theta_r}}$ with the compatible function approximation error. We will also invoke Cauchy-Schwartz inequality for vectors $\sum_i a_ib_i \leq \sqrt{\sum_i a_i^2}\cdot\sqrt{\sum_i b_i^2}$ and random variables $\mathbb{E}|XY| \leq \sqrt{\mathbb{E}X^2}\sqrt{\mathbb{E}Y^2}$.

Conditioned on all the parameters θ_r, ω_r, we observe that

$$
\begin{aligned}
& U\left(\pi^{\theta_r}; \pi^\star\right) - U\left(\pi^{\theta_{r+1}}; \pi^\star\right) \\
=& \mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \left[D_{\mathrm{KL}}\left(\pi_h^\star(\cdot \mid Z_h) \| \pi_h^{\theta_r}(\cdot \mid Z_h)\right) - D_{\mathrm{KL}}\left(\pi_h^\star(\cdot \mid Z_h) \| \pi_h^{\theta_{r+1}}(\cdot \mid Z_h)\right)\right] \\
=& \mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \int_{\widehat{\mathsf{A}}} dA_h \cdot \pi_h^\star(A_h|Z_h) \left(\ln \pi_h^{\theta_{r+1}}(A_h|Z_h) - \ln \pi_h^{\theta_r}(A_h|Z_h)\right) \\
\geq& \mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \left((\theta_{r+1} - \theta_r) \nabla_\theta \ln \pi_h^{\theta_r}(A_h|Z_h) - \frac{\beta}{2} \|\theta_r - \theta_{r+1}\|_2^2 \right) \quad //\beta\text{-smoothness} \\
=& \eta_r \cdot \mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \left(\omega_r^\top \nabla_\theta \ln \pi_h^{\theta_r}(A_h|Z_h) - \widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right)\right) + \eta_r \cdot \mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right) \\
& - \mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \frac{\beta}{2} \eta_r^2 \|\omega_r\|_2^2 \quad //\text{Telescope the advantage function and invoke } \theta_{r+1} = \theta_r + \eta_r \omega_r \\
=& -\eta_r \cdot \epsilon^\star(\omega_r; \theta_r) + \eta_r \left(\mathsf{V}^{\pi^\star} - \mathsf{V}^{\pi^{\theta_r}} \right) - \frac{\beta W^2 H}{2} \eta_r^2 \quad //\text{Performance difference lemma 12}
\end{aligned}
\tag{28}
$$

Here, the Performance difference lemma for POMDPs refers to

Lemma 12 (Performance Difference Lemma Cayci and Eryilmaz (2024))**.** For any policy π', π,

$$
\mathsf{V}^{\pi'} - \mathsf{V}^{\pi} = \mathbb{E}^{\pi'} \left[\sum_{h=0}^{H-1} \gamma^h \ \mathsf{A}_h^{\pi}(Z_h, A_h) \right]
$$

Dividing both sides of Eq. (28) with η_r, we obtain the following fact when conditioned on θ_r and ω_r:

$$
\mathsf{V}^{\pi^\star} - \mathsf{V}^{\pi^{\theta_r}} \leq \frac{U(\pi^{\theta_r}; \pi^\star) - U(\pi^{\theta_{r+1}}; \pi^\star)}{\eta_r} + \epsilon^\star(\omega_r; \theta_r) + \frac{\beta W^2 H}{2} \eta_r \tag{29}
$$

Fix $\eta_r \equiv \eta$ for all r. Conditioned on all the parameters θ_r, ω_r returned by SGD, we take the average over all r and obtain

$$
\begin{aligned}
\frac{1}{R} \sum_{r=0}^{R-1} \mathsf{V}^{\pi^\star} - \mathsf{V}^{\pi^{\theta_r}} \leq& \frac{1}{\eta R} \sum_{r=0}^{R-1} U(\pi^{\theta_r}; \pi^\star) - U(\pi^{\theta_{r+1}}; \pi^\star) + \frac{1}{R} \sum_{r=0}^{R-1} \epsilon^\star(\omega; \theta_r) + \frac{\beta W^2 H}{2} \eta \\
=& \frac{\mathbb{E}^{\pi^\star} \sum_{h=0}^{H-1} \gamma^h \left[D_{\mathrm{KL}}\left(\pi_h^\star(\cdot \mid Z_h) \| \pi_h^0(\cdot \mid Z_h)\right) - D_{\mathrm{KL}}\left(\pi_h^\star(\cdot \mid Z_h) \| \pi_h^R(\cdot \mid Z_h)\right)\right]}{\eta R} \\
& + \frac{1}{R} \sum_{r=1}^{R} \epsilon^\star(\omega; \theta_r) + \frac{\beta W^2 H}{2} \eta \quad //U \text{ is non-negtive} \\
\leq& \frac{HD}{\eta R} + \frac{\beta W^2 H}{2} \eta + \frac{1}{R} \sum_{r=0}^{R-1} \epsilon^\star(\omega_r; \theta_r) \quad // \text{ Definition 4.} \\
\leq& HW \sqrt{\frac{2\beta D}{R}} + \frac{1}{R} \sum_{r=0}^{R-1} \epsilon^\star(\omega_r; \theta_r) \quad //\text{Optimize } \eta_r \text{ over } \mathbb{R} \text{ to get } \eta = \frac{\sqrt{2D/\beta}}{W} \frac{1}{\sqrt{R}}
\end{aligned}
\tag{30}
$$

Next, we connect the term $\epsilon^\star(\omega; \theta_r)$ with the compatible function approximation error introduced in Eq. (25):

$$
\begin{aligned}
&\epsilon^\star(\omega_r;\theta_r)\\
:=&\mathbb{E}^{\pi^\star}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right)-{\omega_r}^\top\nabla_\theta\ln\pi_h^{\theta_r}\left(A_h\mid Z_h\right)\right)\quad //\text{Definition in Eq. (26)}\\
=&\mathbb{E}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^{\frac{h}{2}}\frac{d\mathbb{P}^{\pi^\star}}{d\mathbb{P}^{\pi^{\theta_r}}}\gamma^{\frac{h}{2}}\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right)-{\omega_r}^\top\nabla_\theta\ln\pi_h^{\theta_r}\left(A_h\mid Z_h\right)\right)\\
\le&\mathbb{E}^{\pi^{\theta_r}}\sqrt{\sum_{h=0}^{H-1}\gamma^h\left(\frac{d\mathbb{P}^{\pi^\star}}{d\mathbb{P}^{\pi^{\theta_r}}}\right)^2}\sqrt{\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right)-{\omega_r}^\top\nabla_\theta\ln\pi_h^{\theta_r}\left(A_h\mid Z_h\right)\right)^2}\\
&//\text{Cauchy-Schwartz for vectors}\\
\le&\sqrt{\mathbb{E}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\frac{d\mathbb{P}^{\pi^\star}}{d\mathbb{P}^{\pi^{\theta_r}}}\right)^2}\sqrt{\mathbb{E}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right)-{\omega_r}^\top\nabla_\theta\ln\pi_h^{\theta_r}\left(A_h\mid Z_h\right)\right)^2}\\
&//\text{Cauchy-Schwartz for random variables}\\
=&\sqrt{\mathbb{E}^{\pi^\star}\frac{d\mathbb{P}^{\pi^\star}}{d\mathbb{P}^{\pi^{\theta_r}}}\sum_{h=0}^{H-1}\gamma^h}\sqrt{\mathbb{E}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}\left(\bar{Z}_h\right)-{\omega_r}^\top\nabla_\theta\ln\pi_h^{\theta_r}\left(A_h\mid Z_h\right)\right)^2}\\
\le&\sqrt{\kappa_r H}\cdot\sqrt{\mathcal{L}_{\widehat{\mathsf{A}}}\left(\omega_r;\theta_r,\pi^{\theta_r}\right)}\quad //\text{Definition of }\kappa_r\text{ in Assumption (9) and }\mathcal{L}_{\widehat{\mathsf{A}}}\text{ in Eq. (25).}
\end{aligned}
\tag{31}
$$

Taking the expectation w.r.t. $\theta_{0:R-1}$ and $\omega_{0:R-1}$, we obtain

$$
\begin{aligned}
&\frac{1}{R}\sum_{r=0}^{R-1}\mathbb{E}\ \epsilon^\star(\omega_r;\theta_r)\\
\le&\frac{\sum_{r=0}^{R-1}}{R}\sqrt{H\mathbb{E}\ \kappa_r}\sqrt{\mathbb{E}_{\omega_r,\theta_r}\mathcal{L}_{\widehat{\mathsf{A}}}(\omega_r;\theta_r,\pi^{\theta_r})}\quad //\text{Cauchy-Schwartz for random variables}
\end{aligned}
\tag{32}
$$

$$
\begin{aligned}
\le&\sqrt{H}\sqrt{\frac{\sum_{r=0}^{R-1}}{R}\mathbb{E}\ \kappa_r}\cdot\sqrt{\frac{\sum_{r=0}^{R-1}}{R}\mathbb{E}\ \mathcal{L}_{\widehat{\mathsf{A}}}\left(\omega_r;\theta_r,\pi^{\theta_r}\right)}\quad //\text{Cauchy-Schwartz for vectors}\\
=&\sqrt{\kappa H}\cdot\sqrt{\frac{\sum_{r=0}^{R-1}}{R}\mathbb{E}\ \mathcal{L}_{\widehat{\mathsf{A}}}\left(\omega_r;\theta_r,\pi^{\theta_r}\right)}\quad //\text{Definition of }\kappa\text{ in Eq. (9)}
\end{aligned}
\tag{33}
$$

Bring Eq. (33) to Eq. (30), we conclude the proof. □

Proof of Step 2. Recall that under softmax parameterizaton, $\pi_h^\theta(a_h|z_h) := \frac{\exp f_\theta(z_h,a_h;h)}{\sum_{a_h\in\mathcal{A}}\exp f_\theta(z_h,a_h;h)}$ Consequently, the log-likelihood of the policy distribution is

$$
\log\pi_h^\theta(a_h|z_h) = f_\theta(z_h,a_h;h) - \log\sum_{a_h\in\mathcal{A}}\exp f_\theta(z_h,a_h;h)
$$

Taking the derivative to both sides, the score function of softmax policies is

$$
\nabla_\theta\log\pi_h^\theta(a_h|z_h) = \nabla_\theta f_\theta(z_h,a_h;h) - \mathbb{E}_{a_h\sim\pi_h^\theta(\cdot|z_h)}\nabla_\theta f_\theta(z_h,a_h;h) \tag{34}
$$

which can be viewed as the centered value of the un-normalized output of the neural network f_θ. Interestingly, this structure mirrors the relationship between the advantage function and the action-value function

$$
\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}(\bar{Z}_h) = \widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h) - \mathbb{E}_{A_h\sim\pi^\theta(\cdot|Z_h)}\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(Z_h,A_h)\quad a.s. \tag{35}
$$

With these facts, we immediately conclude that the compatible function approximation error $\mathcal{L}_{\widehat{\mathsf{A}}}(\omega_r;\theta_r;\pi^{\theta_r})$ defined in Eq. (25) is dominated by the Q-compatible function approximation error $\mathcal{L}_{\widehat{\mathsf{Q}}}(\omega_r;\theta_r;\pi^{\theta_r})$ introduced

in Eq. (7), under neural softmax parameterization.

$$
\begin{aligned}
&\mathcal{L}_{\widehat{\mathsf{A}}}(\omega_r;\theta_r;\pi^{\theta_r})\\
=&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{A}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta\ln\pi_h^{\theta_r}(A_h\mid Z_h)\right)^2 \quad //\text{Definition in Eq. (25)}\\
=&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\Big(\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\mathbb{E}_{A_h\sim\pi^\theta(\cdot|Z_h)}\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(Z_h,A_h)\\
&-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)+\omega_r^\top\mathbb{E}_{A_h\sim\pi_h^\theta(\cdot|Z_h)}\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\Big)^2 \quad //\text{Eq. (34) and Eq. (35)}\\
=&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)-\mathbb{E}_{A_h\sim\pi^\theta(\cdot|Z_h)}\left(\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(Z_h,A_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)\right)^2\\
=&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\,\mathrm{Var}_{A_h\sim\pi_h^{\theta_r}(\cdot|Z_h)}\left[\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right]\\
\le&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\mathbb{E}_{A_h\sim\pi_h^{\theta_r}(\cdot|Z_h)}\left(\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)^2 \quad //\text{Variance is less than second moment.}\\
=&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathsf{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)^2=\mathcal{L}_{\widehat{\mathsf{Q}}}(\omega_r;\theta_r;\pi^{\theta_r}) \quad //\text{Definition in Eq. (7)}
\end{aligned}
$$

□

8.2.2 Proof of Lemma 7

We will show $\mathcal{L}_{\widehat{\mathsf{Q}}}(\omega;\theta,\pi^\theta)\le 2\cdot\epsilon_{\text{stat}}(\omega;\theta)+2\cdot\epsilon_{\text{approx}}(\theta)+4\cdot\mathcal{E}^{\pi^\theta}\left(\widehat{\psi}\right)$

Proof. We use several telescoping technique to associate the last display with the evaluation error. In the forthcoming derivations, we will write $\widehat{Q}_h^{\pi^{\theta_r}}$ as the approximated Q-function, and refer to the corresponding Bellman error

$$
\mathcal{E}^{\pi^{\theta_r}}(\psi):=\frac{1}{2}\mathbb{E}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\mathsf{Q}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)\right)^2 \tag{36}
$$

as the evaluation error $\epsilon_{\text{eval}}(\theta_r)$. We will also invoke the definitions of the statistical and approximation error for Q-NPG algorithm, which we would like to repeat it here for the reader's convenience.

$$
\epsilon_{\text{approx}}(\theta_r):=\inf_{\omega\in\mathbb{R}^{d_\theta}}\ \mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)^2 \tag{37}
$$

$$
\epsilon_{\text{stat}}(\omega_r;\theta_r):=\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-{\omega_r}^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)^2-\epsilon_{\text{approx}}(\theta_r) \tag{38}
$$

Now we are ready to state that,

$$
\begin{aligned}
&\mathcal{L}_{\mathsf{Q}}(\omega_r;\theta_r;\pi^{\theta_r})=\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\mathsf{Q}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)^2\\
=&\mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}}\sum_{h=0}^{H-1}\gamma^h\left(\mathsf{Q}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)+\widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h)-\omega_r^\top\nabla_\theta f_{\theta_r}(Z_h,A_h;h)\right)^2 \quad //\text{Telescope a term of } \widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}
\end{aligned}
$$

$$
\begin{aligned}
&\leq 2 \cdot \mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}} \sum_{h=0}^{H-1} \gamma^h \left(\mathsf{Q}_h^{\pi^{\theta_r}}(\bar{Z}_h) - \widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h) \right)^2 + 2 \cdot \mathbb{E}_{\mathcal{P}}^{\pi^{\theta_r}} \sum_{h=0}^{H-1} \gamma^h \left(\widehat{\mathcal{Q}}_h^{\pi^{\theta_r}}(\bar{Z}_h) - \omega_r^\top \nabla_\theta f_{\theta_r}(Z_h, A_h; h) \right)^2 \\
&\quad //(a+b)^2 \leq 2a^2 + 2b^2 \\
&= 4 \cdot \epsilon_{\text{eval}}(\theta_r) + 2 \cdot (\epsilon_{\text{stat}}(\omega_r; \theta_r) + \epsilon_{\text{approx}}(\theta_r)) \quad //\text{Eq. (36), Eq. (37), and Eq. (38).}
\end{aligned}
$$

Putting things together, we conclude

$$
\mathcal{L}_{\widehat{\mathsf{Q}}}(\omega_r; \theta_r; \pi^{\theta_r}) \leq 4\epsilon_{\text{eval}}(\theta_r) + 2\epsilon_{\text{stat}}(\omega_r; \theta_r) + 2\epsilon_{\text{approx}}(\theta_r) \tag{39}
$$

□

Remark 8. Even though the compatible function approximation error relates to the advantage function, we don't need to calculate the advantage function directly from the Q-functions using $\mathsf{A}_h^{\pi^{\theta_r}}(\bar{Z}_h) = \mathsf{Q}_h^{\pi^{\theta_r}}(\bar{Z}_h) - \mathbb{E}_{A_h \sim \pi^\theta(\cdot|Z_h)} \mathsf{Q}_h^{\pi^{\theta_r}}(Z_h, A_h)$, as done in (Cayci and Eryilmaz, 2024). In fact, this conversion is possibly intractable, since it requires evaluating all possible actions for every historical input, which results in a high computational cost of $O\left(\mathcal{O}^H \mathcal{A}^H\right)$. To address this challenge, we can directly adopt a Q-function estimate and run Q-NPG instead, which still allows us to control the error of $\mathcal{L}_{\mathsf{A}}(\omega_r; \theta_r; \pi^{\pi^{\theta_r}})$, as shown in Eq. (39).

8.3 Proof of Lemma 1

Lemma 1 is a consequence of the following result, which characterizes the bound on the Bellman error:

Lemma 13 (Bellman Error Bound for Algorithm 2)**.** *Set $\iota_r = \frac{a}{\mu_\phi(1-\gamma)(r+a)}$ where a is any constant larger than $\frac{9}{2}$. For any local learning rate $\alpha_r \leq \frac{\mu_\phi(1-\gamma)}{16T}$ and global learning rate defined via $\eta_r := \frac{\iota_r}{\alpha_r K}$, the Bellman error of Algorithm 2's output parameter is controlled by the following upper-bound*

$$
\mathbb{E}\, \mathcal{E}\left(\widehat{\boldsymbol{\psi}}^R\right) \leq \widetilde{O}\left(\frac{H d_Q}{R^3} + \frac{1}{{\nu_\phi}^2} \left[\min\left\{ \frac{1}{{\nu_\phi}^2 T^2}, \frac{1}{R^2} \right\} G^2 \; + \; \left(\min\left\{ \frac{1}{{\nu_\phi}^2 K T^2}, \frac{1}{TR} \right\} + \frac{1}{MT} \right) H \sigma_g^2 \right] \right) \tag{40}
$$

if we do stochastic averaging with weights $w_r \propto (r+2)^2$ after $R_0 = \frac{128a}{\nu_\phi^2}$ rounds of synchronizations. σ_g^2 and G are specified in Definitions 2 and 1. Here, we omit the policy parameter in the expression of the Bellman error, for brevity. This result holds for all policy π^θ.

We will show Lemma 13 in Appendix 8.5. It builds upon Lemmas 3, 4, 5, and 14, whose proofs are provided in Appendix 8.1.1, 8.1.2, 8.1.3, and 8.4, respectively.

Given Lemma 13, in what follows, we design a parallel training setup to make the error bound in Lemma 13 be dominated by the varince reduction term, so that we will benefit from parallelism via a linear speedup in sample complexity. This leads to the proof of Lemma 1.

Proof of Lemma 1. Simple calculation reveals that Eq. (40) reduces to $\mathbb{E}\, \mathcal{E}\left(\widehat{\psi}^R\right) \leq \widetilde{O}\left(\frac{\frac{1}{{\nu_\phi}^2} H \sigma_g^2 + H d}{MT} \right)$ under the following conditions

$$
K \leq \frac{T^{2/3}}{M^{1/3}}, \quad \frac{\Gamma(K; \nu_\phi)}{K} < \frac{T}{M}, \quad \text{and} \quad \Gamma(K; \nu_\phi) < \frac{H \sigma_g^2 T}{G^2 M} \tag{41}
$$

The sample complexity to obtain an ϵ-accurate Q-function estimate is $N_{\text{eval}} \leq \widetilde{O}\left(\frac{1}{M\epsilon} \left(\frac{H\sigma_g^2}{{\nu_\phi}^2} + H d_Q \right) \right)$ trajectories, or $\widetilde{O}\left(\frac{H}{M\epsilon} \left(\frac{H\sigma_g^2}{{\nu_\phi}^2} + H d_Q \right) \right)$ observation-action tuples.

To satisfy the constraints in Eq. (41), we can select the communication interval K according to the ill-condition number ν_ϕ, using the analysis results in Section 8.1.3:

- When ν_ϕ is small, meaning that the feature matrix is ill-defined, we need to carefully limit the number of local steps to suppress local drift. We recommend choosing local steps

$$K \leq \min\left\{ \frac{T^{2/3}}{M^{1/3}} \ , \ \frac{2}{e}\frac{T}{M} \ , \ \sqrt{\frac{2}{e}\cdot\frac{H\sigma_g^2}{G}\cdot\frac{T}{M}}, \right\} \tag{42}$$

so that the constraints in Eq. (41) is met using the fact that $\Gamma(K;\nu_\phi) \leq \frac{eK^2}{2}$. Moreover, in this scenario the error bound in Eq. (51) is controlled via

$$\mathbb{E}\ \mathcal{E}\left(\widehat{\psi}^R\right) \leq \widetilde{O}\left(\frac{Hd^2}{MT} + \frac{1}{{\nu_\phi}^2}\left[\left(\frac{1}{MT}+\frac{1}{RT}\right)H\sigma_g^2 + \frac{1}{R^2}G^2\right]\right) \leq \widetilde{O}\left(\frac{Hd}{MT} + \frac{H\sigma_g^2}{{\nu_\phi}^2 MT}\right) \tag{43}$$

and the communication complexity is

$$R = \frac{T}{K} \geq \max\left\{ (MT)^{1/3} \ , \ \frac{e}{2}M \ , \ \sqrt{\frac{e}{2}\frac{G}{H\sigma_g^2}MT} \right\} \tag{44}$$

- When ν_ϕ is large, it implies that the feature matrix is well-defined, we need to ensure that the total and local compuation budgets are large enough to keep the last two terms in Eq. (51) quickly converges. When that happends, the variance reduction term will stands out of our bound, which exhibits the benefits of parallelism by a linear speedup. To this end, we discover that a) $K \leq \frac{T^{2/3}}{M^{1/3}}$ implies that $\frac{1}{R^3} \leq \frac{1}{MT}$, b) $K \geq \frac{1}{{\nu_\phi}^2}\frac{M}{T}$ implies that $\frac{\Gamma(K;\nu_\phi)}{KT^2} \leq \frac{e}{{\nu_\phi}^2 KT^2} \lesssim \frac{1}{MT}$ and c) $\frac{T}{M} \geq \frac{1}{{\nu_\phi}^2}\frac{G^2}{H\sigma_g^2}$ impliese that $\frac{\Gamma(K;\nu_\phi)}{T^2}G^2 \leq \frac{e}{{\nu_\phi}^2 T^2}G^2 \lesssim \frac{H\sigma_g^2}{MT}$. When $a)$ to $c)$ are simultaneously satisfied, the sample complexity will scale with $\frac{1}{MT}$. One possible scenarios is to pick $K = \frac{T^{2/3}}{M^{1/3}}$ when $\frac{T}{M} \geq \frac{1}{{\nu_\phi}^2}\frac{G^2}{H\sigma_g^2}$. This choice directly meets the requirements of a) and c), while also ensures b) $\frac{T}{M} \geq \frac{1}{{\nu_\phi}^2 K}$ since second-moment is larger than the variance. Under this setup,

$$\mathbb{E}\ \mathcal{E}\left(\widehat{\psi}^R\right) \leq \widetilde{O}\left(\frac{Hd^2}{MT} + \frac{1}{{\nu_\phi}^2}\left[\left(\frac{1}{MT}+\frac{1}{{\nu_\phi}^2 KT^2}\right)H\sigma_g^2 + \frac{1}{{\nu_\phi}^2 T^2}G^2\right]\right) \leq \widetilde{O}\left(\frac{H\sigma_g^2 + Hd}{{\nu_\phi}^2 MT}\right) \tag{45}$$

where the number of communications $R = \frac{T}{K} = (MT)^{1/3}$ improves Eq. (44), at the cost of a slightly increase Bellman error.

We derive the sample complexity of Algorithm 2 from Eqs (43) and (45), by solving the minumum T that ensures $\mathbb{E}\ \mathcal{E}\left(\widehat{\psi}^R\right) \leq \epsilon$. The communication complexity naturally follows from the relation $T = RK$. □

Remark 9. Additional to the Bellman error bound in Lemma 13, the proofs to the complexity of the Temporal Difference Learning algorithm (Lemma 2) and the Actor-Critic Algorithm (Corollary 1) are deferred to Appendix 9.2 and 9.3, respectively.

Next, we present a series of lemmas (Lemmas 3, 4, 5, and 14) that form the building block of Lemma 13. Then, we prove Lemma 13 in Appendix 8.5.

8.4 Proof of Lemma 14

Building on previous results, we can characterize the relationship between the potential function and the Bellman error in the following lemma.

Lemma 14. *The potential function and the Bellman error possess the following relation:*

$$U\left(\widetilde{\psi}^{r+1}\right) \leq \left(1 - \frac{\iota_r}{2}\nu_\phi\right)U\left(\widetilde{\psi}^r\right) - \frac{\iota_r}{2}\nu_\phi \mathbb{E}\ \mathcal{E}\left(\widetilde{\psi}^r\right) + \frac{9}{4}\Gamma(K;\nu_\phi)\left(\frac{H\sigma_g^2}{RK^3} + \frac{2G^2}{RK^2}\right)\iota_r^2 + \frac{2H\sigma_g^2}{MK}\iota_r^2$$

Proof. First we invoke a standard result in federated RL that links the gradient-dependent terms in Lemma 4 and Lemma 3. Readers may refer to Eq.(a) in the proofs of Proposition 4 and the second in equality for Lemma 10 in (Wang et al., 2023) for details. With Definition 1 and Lemma 10, we have

$$\mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star, \mathbb{E}_z g\left(\widetilde{\psi}^r; Z\right)\right\rangle + \eta_r\alpha_r K \cdot 2\left\|\mathbb{E}_{\bar{Z}}^{\pi}\, g\left(\widetilde{\psi}^r; \bar{Z}\right)\right\|_2^2 \leq -\mu_\phi(1-\gamma)U\left(\psi^r\right) + 16\mathbb{E}\,\mathcal{E}\left(\widetilde{\psi}^r\right)$$

Taking Lemma 3 and Lemma 4 to Eq. (10),

$$\begin{aligned} U(\widetilde{\psi}^{r+1}) =& U(\widetilde{\psi}^r) + \underbrace{2\mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star\ ,\ \widetilde{\psi}^{r+1} - \widetilde{\psi}^r\right\rangle}_{\text{Progression Direction}} + \underbrace{\mathbb{E}\left\|\widetilde{\psi}^{r+1} - \widetilde{\psi}^r\right\|_2^2}_{\text{Progression Distance}} \\ \leq& \left(1 - \frac{\eta_r\alpha_r K}{c}\right)U(\widetilde{\psi}^r) + \eta_r\alpha_r K \cdot \left[\left(-2(1-\gamma)\mu_\phi + \frac{2}{c}\right)\cdot U\left(\psi^r\right) + 64\eta_r\alpha_r K\mathbb{E}\mathcal{E}\left(\widetilde{\psi}^r\right)\right] \\ &+ \eta_r\alpha_r K\left(c + 4\eta_r\alpha_r K\right)\cdot \underbrace{16\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}} + 2(\eta_r\alpha_r K)^2\frac{H\sigma_g^2}{MK} \end{aligned} \tag{46}$$

Next, we further simplify the two terms in the middle, at the cost of several restrictions on the learning rates. First we would like to simplify the second term. With Eq. (20), we discover that as long as the composite learning rates satisfy $\eta_r\alpha_r K \leq \frac{\mu_\phi(1-\gamma)}{128}$, we are able to do suppress the Bellman error with the potential function, under the setting $c = \frac{2}{\mu_\phi(1-\gamma)}$:

$$-\mu_\phi(1-\gamma)\cdot U\left(\psi^r\right) + 64\cdot\eta_r\alpha_r K\mathbb{E}\mathcal{E}\left(\widetilde{\psi}^r\right) \leq \frac{-\mu_\phi(1-\gamma)}{2}\mathbb{E}\mathcal{E}\left(\widetilde{\psi}^r\right) \tag{47}$$

Under the requirement of $\alpha_r < \frac{\mu_\phi(1-\gamma)}{16}$, Lemma 5) suggest that the local drift term is dominated by

$$\underbrace{16\frac{1}{MK}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\psi^{r,k,m} - \widetilde{\psi}^r\right\|_2^2\right]}_{\text{Local drift}} \leq 16\Gamma(K;\nu_\phi)\left(\frac{H\sigma_g^2}{K} + 2G^2\right)\cdot\alpha_r^2$$

where $\Gamma(K;\nu_\phi)$ is defined in Eq. (21). We decrease the local learning rate with a factor of H to offset the non-stationarity of POMDPs.
With these two simplifications, we arrive at

$$\begin{aligned} U(\widetilde{\psi}^{r+1}) =& U(\widetilde{\psi}^r) + \underbrace{2\mathbb{E}\left\langle \widetilde{\psi}^r - \psi^\star\ ,\ \widetilde{\psi}^{r+1} - \widetilde{\psi}^r\right\rangle}_{\text{Progression Direction}} + \underbrace{\mathbb{E}\left\|\widetilde{\psi}^{r+1} - \widetilde{\psi}^r\right\|_2^2}_{\text{Progression Distance}} \\ \leq& \left(1 - \frac{\eta_r\alpha_r K}{2}\nu_\phi\right)U(\widetilde{\psi}^r) - \eta_r\alpha_r K\cdot\frac{\nu_\phi}{2H}\mathbb{E}\,\mathcal{E}\left(\widetilde{\psi}^r\right) \\ &+ (\eta_r\alpha_r K)^2\left(\frac{2}{\alpha_r K\nu_\phi} + 4\right)\cdot 16\Gamma(K;\nu_\phi)\left(\frac{H\sigma_g^2}{K} + 2G^2\right)\cdot\alpha_r^2 \\ &+ 2(\eta_r\alpha_r K)^2\frac{H\sigma_g^2}{MK} \end{aligned} \tag{48}$$

under the condition that $c = \frac{2}{\mu_\phi(1-\gamma)}$, $\nu_\phi = \mu_\phi(1-\gamma)$, $\alpha_r \leq \frac{\mu_\phi(1-\gamma)}{16}$ and $\eta_r\alpha_r K \leq \frac{\mu_\phi(1-\gamma)}{128}$. If further require that $\alpha_r \leq \frac{\mu_\phi(1-\gamma)}{16RK}$, then we could obtain $\left(\frac{2}{\alpha_r K\nu_\phi} + 4\right)\alpha_r^2 \leq \frac{9}{64RK^2}$ and arrive at

$$U\left(\widetilde{\psi}^{r+1}\right) \leq \left(1 - \frac{\iota_r}{2}\nu_\phi\right)U\left(\widetilde{\psi}^r\right) - \frac{\iota_r}{2}\nu_\phi\mathbb{E}\,\mathcal{E}\left(\widetilde{\psi}^r\right) + \frac{9}{4}\Gamma(K;\nu_\phi)\left(\frac{H\sigma_g^2}{RK^3} + \frac{2G^2}{RK^2}\right)\iota_r^2 + \frac{2H\sigma_g^2}{MK}\iota_r^2$$

□

Finally, we show Lemma 13.

8.5 Proof of Lemma 13

Proof. Without loss of generality, we require the feature vectors to be normalized 2-norms, so that the norm of the parameters and the Bellman error defined in Eq. (3)) have the following (crude) upper bounds

$$\|\psi\|_2 \leq \sqrt{Hd}\ , \quad \mathcal{E}(\psi) \leq \|\psi - \psi^\star\|_2 \tag{49}$$

Readers may refer to Lemma B.2 of (Efroni et al., 2022) or Section 2.1 in (Jin et al., 2020b) for a reference.

In what follows, we write $\eta_r \alpha_r K$ as ι_r, $\frac{\nu_\phi}{2}\mathbb{E}\,\mathcal{E}(\widetilde{\psi}^r)$ as e_r, $U(\widetilde{\psi}^r)$ as u_r and A as $\frac{9}{4}\Gamma(K;\nu_\phi)\left(\frac{H\sigma_g^2}{RK^3} + \frac{2G^2}{RK^2}\right) + \frac{2H\sigma_g^2}{MK}$, to further abbreviate expressions in Lemma 14. Under the constraints that $\alpha_r \leq \frac{\mu_\phi(1-\gamma)}{16RK}$, $\eta_r\alpha_r K = \iota_r \leq \frac{\mu_\phi(1-\gamma)}{128}$, we obtain

$$e_r \leq \left(\frac{1}{\iota_r} - \frac{\nu_\phi}{2}\right) u_r - \frac{1}{\iota_r} u_{r+1} + A\iota_r \tag{50}$$

Now we set the stochastic averaging weight as $w_r = (r+2)^2$, total weight as $W = \sum_{r=0}^{R} w_r \geq R^3/3$, composite learning rate as $\iota_r = \frac{a}{\mu_\phi(1-\gamma)(r+a)}$ where a is any positive constant larger than $\frac{9}{2}$. Then when the total rounds of communication is sufficiently large such $\forall R \geq R_0$, we ensure that $\iota_r \leq \frac{\nu_\phi}{128}$, the weighted average of the last error terms is controlled by

$$\begin{aligned}
\frac{1}{W}\sum_{r=R_0}^{R} w_r e_r \leq & \frac{3\nu_\phi}{R^3}\left(1 - \frac{1}{a}\right) u_0 + \frac{3a}{\nu_\phi}\left(\frac{1}{2R} + (a-2)^2 \frac{\ln(1 + \frac{R+1}{a})}{R^3}\right) A \\
= & \widetilde{O}\left(\frac{\nu_\phi}{R^3}\left\|\bar{\psi}^0 - \psi^\star\right\|_2^2 + \frac{H\sigma_g^2}{\nu_\phi MKR} + \Gamma(K;\nu_\phi)\frac{H\sigma_g^2}{\nu_\phi R^2 K^3} + \Gamma(K;\nu_\phi)\frac{G^2}{\nu_\phi R^2 K^2}\right) \\
\leq & \widetilde{O}\left(\frac{\nu_\phi Hd}{R^3} + \frac{H\sigma_g^2}{\nu_\phi MKR} + \Gamma(K;\nu_\phi)\frac{H\sigma_g^2}{\nu_\phi R^2 K^3} + \Gamma(K;\nu_\phi)\frac{G^2}{\nu_\phi R^2 K^2}\right)
\end{aligned}$$

The proofs of the first inequality is similar to Lemma 3.4 in (Stich, 2019) and we omit it for brevity. The last display is due to Eq. (49). By the convexity of the Bellman error, we conclude:

$$\mathbb{E}\,\mathcal{E}\left(\widehat{\psi}^R\right) \leq \widetilde{O}\left(\frac{Hd}{R^3} + \frac{1}{{\nu_\phi}^2}\left[\left(\frac{1}{MT} + \frac{\Gamma(K;\nu_\phi)}{KT^2}\right) H\sigma_g^2 + \frac{\Gamma(K;\nu_\phi)}{T^2} G^2\right]\right) \tag{51}$$

where we write the total number of gradient steps or sample trajectories taken by each parallel machine as $T = RK$. We remind the readers that the error bound holds true when $\alpha_r \leq \frac{\mu_\phi(1-\gamma)}{16RK}$, $\eta_r = \frac{\iota_r}{\alpha_r K}$, $\iota_r = \frac{a}{\mu_\phi(1-\gamma)(r+a)}$, $w_r = (r+2)^2$ and we take the weighted average after $r \geq \frac{128a}{{\nu_\phi}^2}$. We complete the proof by the fact that $\Gamma(K;\nu_\phi) \leq O\left(\min\left\{\frac{1}{{\nu_\phi}^2 KT^2}, K^2\right\}\right)$ and $T = KR$. □

Remark 10. In Lemma 13, the coefficients $\min\left\{\frac{1}{{\nu_\phi}^2 KT^2}, \frac{1}{TR}\right\}$ and $\min\left\{\frac{1}{{\nu_\phi}^2 T^2}, \frac{1}{R^2}\right\}$ are novel ingredients compared with previous works such as (Wang et al., 2023; Woo et al., 2024), owning to a precise characterization on the drift terms in Lemma 5.

9 Auxiliary Results

In this section, we provide some remark on the lemmas and theorem introduced in the previous sections. Then, we prove several corollaries of these lemmas concerning with the complexity of our algorithms, and include the reference to a technical result used in the proof of Theorem 1.

9.1 Remarks on the Natural Policy Gradient Algorithm

9.1.1 The Policy Gradient

First, we generalize the policy gradient theorems (Sutton and Barto, 2018) to POMDPs by the following lemma:

Lemma 15 (Policy Gradient Theorem, Baxter and Bartlett (2001)).

$$\nabla_\theta \mathsf{V}^\pi = \mathbb{E}^{\pi^\theta}\left[\sum_{h=0}^{H-1}\gamma^h \mathsf{Q}_t^{\pi^\theta}(Z_t, A_t)\nabla_\theta \ln \pi_t^\theta(A_t|Z_t)\right] = \mathbb{E}^{\pi^\theta}\left[\sum_{h=0}^{H-1}\gamma^h \mathsf{A}_t^{\pi^\theta}(Z_t, A_t)\nabla_\theta \ln \pi_t^\theta(A_t|Z_t)\right] \tag{52}$$

This lemma will be used to derive the Polyak-Łojasiewicz constant of the compatible function approximation problem, which is later detailed in Remark 12.

9.1.2 Regularity Constant

Remark 11 (Discussion on κ). The authors of (Cayci and Eryilmaz, 2024) introduced a similar concept of κ. In this work, we replace their definition with a finer characterization by investigating how the regularity constant evolves during policy improvement. Recall that we defined κ_r by $\kappa_r = \mathbb{E}^{\pi^\star}\frac{d\mathbb{P}^{\pi^\star}}{d\mathbb{P}^{\pi^{\theta_r}}}$ and referred to κ as $\kappa := \sqrt{\frac{1}{R}\sum_{r=1}^R \mathbb{E}\,\kappa_r}$. By expanding the trajectory probability along the horizon,

$$\begin{aligned}\mathbb{P}^{\pi^\theta}(\bar{Z}_H) =& \rho(S_0)\mathbb{O}_0(O_0|S_0)\pi_0^\theta(A_0|O_0)\mathbb{T}_1(S_1|S_0,A_0)\mathbb{O}_1(O_1|S_1)\pi_1^\theta(A_1|Z_1)\dots \\ & \mathbb{T}_{H-1}(S_{H-1}|S_{H-2},A_{H-2})\mathbb{O}_{H-1}(O_{H-1}|S_{H-1})\pi_{H-1}^\theta(A_{H-1}|Z_{H-1})\end{aligned}$$

we can relate the sample-path distribution $\mathbb{P}^{\pi^\theta}$ with the policy distribution, and conclude that $\kappa_r = \mathbb{E}^{\pi^\star}\prod_{t=0}^{H-1}\frac{\pi_t^\star(A_t|Z_t)}{\pi_t^{\theta_r}(A_t|Z_t)} = \mathbb{E}^{\pi^\star}\exp\sum_{t=0}^{H-1}\ln\pi_t^\star(A_t|Z_t) - \ln\pi_t^{\theta_r}(A_t|Z_t)$. According to the β-smoothness of the policy (Assumption 5), we can derive an upper bound on κ_r, even in continuous action spaces.

$$\kappa_r \leq \mathbb{E}^{\pi^\star}\exp\sum_{t=0}^{H-1}\beta\left\|\theta^\star - \theta_r\right\|_2 = \exp H\beta\left\|\theta^\star-\theta_r\right\|_2 \qquad \kappa \leq \mathbb{E}\sqrt{\frac{1}{R}\sum_{r=1}^R \exp H\beta\left\|\theta^\star-\theta_r\right\|_2} \tag{53}$$

Equation (53) indicates that κ_r is intricately linked to the global convergence of the NPG algorithm, as it is influenced by the distance between the parameters of the r-th round policy and the optimal policy, measured using the Euclidean norm. Although this bound does not account for the structural properties of the policy, it still provides valuable insights into the policy optimization problem for POMDPs.

Specifically, the loose bound in Equation (53) exhibits an exponential dependency on the horizon length. This challenge can be partly alleviated by imposing structural assumptions on the POMDP transition kernels Cai et al. (2022); Liu et al. (2022a) or by applying regularization techniques Cayci et al. (2021). Similar pessimistic bounds also appear in policy gradient theory for using softmax policy for MDPs Li et al. (2021), where the iteration complexity is of the order $\frac{1}{\eta}|\mathcal{S}|^{\Omega\left(\frac{1}{1-\gamma}\right)}$. As the Natural Policy Gradient updates converge towards the optimal solution, κ_r approaches 1. Reducing κ_r below a small constant becomes increasingly difficult as temporal information increases (when H is large) or when the policy class loses smoothness (when β is large). While this upper bound is dependent on H, it remains unaffected by the sizes of the state and observation spaces, demonstrating the algorithm's robust scalability for large-scale POMDP problems.

Moreover, κ_r represents the exponent of the second-order Rényi divergence between optimal and learned sample path probabilities. By linking κ_r with f-divergences (Ay et al., 2017), we can establish upper and lower bounds on κ related to the geometry of the probability simplex. The following result is even independent of policy parameterization and applicable to potentially continuous action spaces.

$$\begin{aligned}\exp\sum_{t=0}^{H-1}\mathbb{E}^{\pi^\star}D_{KL}\left(\pi_t^\star(\cdot|Z_t)||\pi_t^{\theta_r}(\cdot|Z_t)\right) \leq \kappa_r \leq 1+\chi^2\left(\mathbb{P}^{\pi^\star}||\mathbb{P}^{\pi^{\theta_r}}\right) \\ \leq 1 + \left\|\prod_{t=0}^{H-1}\frac{\pi^{\theta^\star}(A_t|Z_t)}{\pi^{\theta_r}(A_t|Z_t)}\right\|_\infty \sqrt{2\sum_{t=0}^{H-1}\mathbb{E}^{\pi^\star}D_{KL}\left(\pi_t^\star(\cdot|Z_t)||\pi_t^{\theta_r}(\cdot|Z_t)\right)}\end{aligned} \tag{54}$$

Here, $\chi^2(\cdot||\cdot)$ is the Chi-square divergence between two probability measures. The proof is a simple application of basic information-theoretic inequalities detailed in (Golowich et al., 2022a,b), and we omit it for brevity. We observe that as the KL-divergence between the policies approaches zero, κ_r converges to 1. It is promising that applying regularization for POMDPs based on information theory can help reduce the upper bound on κ_r, which we leave for future work.

Next, we prove the complexity of the TD learning algorithm (Lemma 2) and the Actor-Critic algorithm (Corollary 1), and introduce Lemma 16, which is a crucial component in the proof of Theorem 1.

9.2 Proof of Lemma 2

Proof. Let us first show the result concerning sample complexity. Under the convention that $T = RK$, if we pick $K = \frac{T^{1/2}}{M^{1/2}}$ (which implies $R = (MT)^{1/2}$), Theorem 1 implies that

$$\mathsf{V}^{\pi^\star} - \mathbb{E}_{\theta_r}\left[\frac{1}{R}\sum_{r=1}^{R}\mathsf{V}^{\pi^{\theta_r}}\right] \leq \frac{HW\sqrt{2\beta D} + \frac{1}{\mu_F}\sqrt{\kappa HL\sigma_w^2}}{\sqrt{MK}} + \epsilon_{\text{app}} = O\left(\frac{1}{\sqrt{MK}}\right) + \epsilon_{\text{app}}$$

where we abbreviated the function approximation error $\frac{\sqrt{\kappa H}}{R}\sum_{r=1}^{R}\sqrt{\mathbb{E}\ \epsilon_{approx}(\theta_r)}$ as ϵ_{app}. Consequently, to ensure ϵ-suboptimality, the number of sample trajectories used in each round r is

$$N_{\text{per_round}} = K \geq \frac{H^2W^2\beta D + \frac{1}{\mu_F}\kappa HL\sigma_w^2}{M(\epsilon - \epsilon_{\text{app}})^2} \approx \frac{H^2W^2\beta D + \frac{1}{\mu_F}\kappa HL\sigma_w^2}{M\epsilon^2}$$

provided that the function approximation error is small. Consequently, the total number of samples used in all the policy updates combined is

$$N_{\text{total}} = T = RK = MK^2 \gtrsim \frac{H^4W^4\beta^2D^2 + \frac{\kappa^2}{\mu_F^2}H^2L^2\sigma_w^2}{M\epsilon^4}$$

, exhibits a linear speedup w.r.t. M. For communication complexity, the number of times of gradient communication equals $R \cdot I = R \cdot O\left(\frac{T^{1/3}}{M^{2/3}}\right) = O\left(\frac{T^{5/6}}{M^{1/6}}\right)$ in each round, which is accompanied with $R = \sqrt{MT}$ times of synchronizations of the policy parameter. □

9.3 Proof of Corollary 1

For the proof of Theorem 1, please refer to Appendix 7. Below we only show the proof of Corollary 1.

According to Eq. (43), the Bellman error of policy evaluation is $\mathbb{E}\,\mathcal{E}^{\pi^\theta}\left(\widehat{\psi}^r\right)$ is controlled by $\widetilde{O}\left(\frac{H}{MT_e}\left(\frac{d_Q+\sigma_g^2}{{\nu_\phi}^2}\right)\right)$. We set the number of local steps for policy improvement $K = \sqrt{\frac{T}{M}}$, number of rounds of policy update $R = \sqrt{MT}$ and the number of steps for policy evaluation as $T_e = K$. This allows the functional suboptimality to be

$$\mathsf{V}^{\pi^\star} - \mathbb{E}_{\theta_{1:R}}\left[\frac{1}{R}\sum_{r=1}^{R}\mathsf{V}^{\pi^{\theta_r}}\right] \lesssim \frac{HW\sqrt{\beta D} + \frac{1}{\mu_F}\sqrt{\kappa HL\sigma_g^2} + \sqrt{\kappa H^2\left(\frac{d_Q+\sigma_g^2}{{\nu_\phi}^2}\right)}}{\sqrt{MK}} + \epsilon_{\text{app}}$$

To ensure ϵ-suboptimality, we need to ensure

$$K \gtrsim \frac{H^2W^2\beta D + \frac{1}{\mu_F^2}\kappa HL\sigma_g^2 + \kappa H^2\left(\frac{d_Q+\sigma_g^2}{{\nu_\phi}^2}\right)}{M(\epsilon - \epsilon_{\text{app}})^2} \approx \frac{H^2W^2\beta D + \frac{1}{\mu_F^2}\kappa HL\sigma_g^2 + \kappa H^2\left(\frac{d_Q+\sigma_g^2}{{\nu_\phi}^2}\right)}{M\epsilon^2}$$

provided that the function approximation error is small. These facts imply that the total sample complexity of Algorithm 1 is

$$N_{\text{total}} = N_{\text{eval}} + N_{\text{improve}} = 2T = 2MK^2 = O\left(\frac{C^2}{M\epsilon^4}\right)$$

where $C = H^2W^2\beta D + \frac{\kappa}{\mu_F^2}HL\sigma_g^2 + \kappa H^2\left(\frac{d_Q+\sigma_g^2}{{\nu_\phi}^2}\right)$.

As for communication complexity, Algorithm 1 requires sending parameter $\theta \in \mathbb{R}^{d_\theta}$ and $\omega \in \mathbb{R}^{d_\theta}$ for $R + C = O\left(\sqrt{MT} + T^{5/6}M^{-1/6}\right)$ times, while communicating Q-network parameter $\psi \in \mathbb{R}^{d_Q}$ for $R \cdot (MT_e)^{1/3} = (MT)^{1/3}$. Putting things together, we need to communicate

$$P_{\text{total}} = O\left((MT)^{1/2} + T^{5/6}M^{-1/6}\right)d_\theta + (MT)^{1/3} \cdot d_Q$$

floating-point parameters.

9.4 Lemma 16

We present a linear speedup guarantee for running a minibatch parallel SGD algorithm on smooth functions under the Poliak-Łojasiewicz condition.

Lemma 16 (Adapted from Theorem 1 of (Haddadpour et al., 2019))**.** *Suppose the following assumptions hold: the stochastic gradients are unbiased and their variance evaluated on a minibatch of size B is bounded as $\mathbb{E}\left[\|\widetilde{\mathbf{g}} - \mathbf{g}\|^2\right] \leq C_1\|\mathbf{g}\|^2 + \frac{\sigma^2}{B}$, where C_1 and σ are non-negative constants, and the objective function $F(\mathbf{x})$ is differentiable and L-smooth: $\|\nabla F(\mathbf{x}) - \nabla F(\mathbf{y})\| \leq L\|\mathbf{x}-\mathbf{y}\|, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, and it satisfies the Polyak-Łojasiewicz (PL) condition with constant $\mu_F : \frac{1}{2}\|\nabla F(\mathbf{x})\|_2^2 \geq \mu_F\left(F(\mathbf{x}) - F\left(\mathbf{x}^*\right)\right), \forall \mathbf{x} \in \mathbb{R}^d$ with $\mathbf{x}^*$ is an optimal solution, that is, $F(\mathbf{x}) \geq F\left(\mathbf{x}^*\right), \forall \mathbf{x}$. For LUPA-SGD (Haddadpour et al., 2019) with τ local updates, if we choose the learning rate as $\eta_t = \frac{4}{\mu_F(t+a)}$ where $a = \alpha\tau + 4$ with α being a constant satisfying $\alpha \exp\left(-\frac{2}{\alpha}\right) < \kappa\sqrt{192\left(\frac{p+1}{p}\right)}$, choose mini-batch size as b, local step number as $K = O\left(\frac{T^{\frac{2}{3}}}{M^{\frac{1}{3}}b^{\frac{1}{3}}}\right)$, the functional suboptimality is controlled by*

$$\mathbb{E}\left[F\left(\bar{\mathbf{x}}_T\right)\right] - F^* \leq \frac{a^3\left(F\left(x_0\right) - F^*\right)}{(T+a)^3} + \frac{4L\sigma^2}{\mu_F^2 bM}\frac{1}{T+a} + \frac{256L^2\sigma^2}{\mu_F^3 bM}\frac{T(K-1)}{(T+a)^3} \lesssim \frac{L}{\mu_F^2}\frac{\sigma^2}{bMT} \tag{55}$$

Remark 12. For linear regression problems where $F(x) = \frac{\|Ax-b\|_2^2}{2}$, it is straightforward to evaluate that it satisfies the PL condition with $\mu_F = \lambda_{\min}\left(A^\top A\right)$. It is also L-smooth, with $L = \lambda_{\max}\left(A^\top A\right)$, where we use $\lambda_{\min}(A)$ and $\lambda_{\max}(B)$ to represent the minumum and maximum eigenvalues of matrix A. Notably, optimizing the compatible function approximation problem is also a least square regression. Specifically, $\underset{\omega}{\text{minimize}}\ \mathcal{L}_{\mathsf{A}}(\omega;\theta;\pi^\theta) = \underset{\omega}{\text{minimize}}\ \omega^\top\mathcal{F}_\theta\ \omega - 2\langle c, \omega\rangle$ where $c = \mathbb{E}^{\pi^\theta}\left[\sum_{h=0}^{H-1}\gamma^h \mathsf{A}^{\pi^\theta}(\bar{Z}_h)\nabla_\theta \ln \pi_h^{\pi^\theta}(A_h|Z_h)\right]$. According to Eq. (52), $c = \nabla_\theta \mathsf{V}^{\pi^\theta}$, which implies that

$$\underset{\omega}{\text{minimize}}\ \mathcal{L}_{\mathsf{A}}(\omega;\theta;\pi^\theta) = \underset{\omega}{\text{minimize}}\ \omega^\top\mathcal{F}_\theta\ \omega - 2\langle\nabla_\theta \mathsf{V}^{\pi^\theta}, \omega\rangle = \underset{\omega}{\text{minimize}}\ \frac{1}{2}\|A\omega - b\|_2^2$$

where A, b is any solution to $A^\top A = 2\mathcal{F}_\theta$ and $b^\top A = \nabla_\theta \mathsf{V}^{\pi^\theta}$. Consequently, the regularity constants for the compatible function approximation problem is $L = 2\lambda_{\max}\left(\mathcal{F}^{\pi^\theta}\right)$ and $\mu_F = 2\lambda_{\min}\left(\mathcal{F}^{\pi^\theta}\right)$.